

A project report on

**Self-Supervised Vision Transformer for
Leukemia Classification from Blood Smear
Images with Explainable Attention Maps**

Submitted in partial fulfillment for the award of the degree of

**Bachelor of Technology in ELECTRONICS
AND COMPUTER ENGINEERING**

by

ABILESH SD (22BLC1169)



VIT[®]

Vellore Institute of Technology

(Deemed to be University under section 3 of UGC Act, 1956)

CHENNAI

SCHOOL OF ELECTRONICS ENGINEERING

November, 2025

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DECLARATION

I hereby declare that the thesis entitled "SELF-SUPERVISED VISION TRANSFORMER FOR LEUKEMIA CLASSIFICATION FROM BLOOD SMEAR IMAGES WITH EXPLAINABLE ATTENTION MAPS" submitted by Abilesh SD (22BLC1169), for the award of the degree of Bachelor of Technology in Electronics and Computer Engineering, Vellore Institute of Technology, Chennai is a record of bonafide work carried out by me under the supervision of Dr. Jani Anbarasi L.

I further declare that the work reported in this thesis has not been submitted and will not be submitted, either in part or in full, for the award of any other degree or diploma in this institute or any other institute or university.

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Date: 3-11-2025

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School of Electronics and Engineering

CERTIFICATE

This is to certify that the report entitled "SELF-SUPERVISED VISION TRANSFORMER FOR LEUKEMIA CLASSIFICATION FROM BLOOD SMEAR IMAGES WITH EXPLAINABLE ATTENTION MAPS" is prepared and submitted by **Abilesh SD (22BLC1169)** to Vellore Institute of Technology, Chennai, in partial fulfillment of the requirement for the award of the degree of **Bachelor of Technology in ELECTRONICS AND COMPUTER ENGINEERING** is a bonafide record carried out under my guidance. The project fulfills the requirements as per the regulations of this University and in my opinion meets the necessary standards for submission. The contents of this report have not been submitted and will not be submitted either in part or in full, for the award of any other degree or diploma and the same is certified.

Signature of the Guide:

Name: Dr. Jani Anbarasi L

Date: 3-11-2025

Signature of the Examiner

Name: M. Priyaadharshini

Date: 5/11/25



Signature of the Examiner

Name: Dr. RAJESH R

Date: 5/11/25

Approved by the Head of Department,
Electronics and Computer Engineering

Name: Dr. Vijayakumar P

Date: 7/11/25

Head of the Department
B.Tech (ECM) Programme
School of Electronics Engineering
Vellore Institute of Technology(VIT)
Vandalur - Kelambakkam Road,
Chennai - 600 127.



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Name: Dr. Jani Anbarasi L

Date:

Signature of the Examiner

Name:

Date:

Signature of the Examiner

Name:

Date:

Approved by the Head of Department,
Electronics and Computer Engineering

Name: Dr. Vijayakumar P

Date:

ABSTRACT

Leukemia is a critical hematological cancer, which is characterized by the atypical growth of immature leukocytes, hence distorting the homeostatic generation of normal hematopoietic cells. The accurate and prompt diagnosis is the key to improving patient outcomes. Traditional diagnostic tools rely on the manual microscopic evaluation of peripheral blood smears, which is a tedious but subjective process with high reliance on the subjective interpretation of the expert. These limitations have triggered the endeavors to develop automated, objective, and reproducible diagnostic systems, which are based on artificial intelligence and deep learning. However, there are still two significant issues: medical data labeled scarcity and the inability to understand the black-box deep learning models.

As a way to reduce these weaknesses, the current paper presents a two-phase deep learning model, which consolidates self-supervised pretraining with supervised fine-tuning that follows, thus, allowing automated classification of leukemia based on blood smear image data. During the first phase, a Masked Autoencoder (MAE) with a Vision Transformer (ViT) architecture is trained on unlabeled microscopic images obtained by the C-NMC Leukemia Classification dataset, and it is trained to learn natural visual representations through reconstruction of stochastically masked image patches. The pretraining enables the encoder to learn the underserved morphological structures without manual annotations. During the secondary step, the already trained encoder is fine-tuned with the help of the labeled subset of the data in order to perform binary classification of normal cells and leukemia blasts where the optimization is aimed at diagnostic discrimination.

Empirical evidence of stable convergence and feature transfer is shown between stages, which highlights the ability of self-supervised learning to significantly improve representation fidelity and reduce the reliance on annotated data. The reliability of the model is assured by quantitative performance measures of accuracy, precision, recall, F1 -score, and ROC -AUC, which support the model in clinical use. The profiles of qualitative reconstructions based on the MAE help to further highlight the ability of the model to internalize the hematological morphology.

To conclude, this study confirms that transformer-based self-supervised learning can be successfully used in medical imaging to detect leukemia. The suggested framework does not only enhance the quality of classification but also enhances transparency and scalability thus making it a promising future step towards interpretable, data-efficient, and clinically reliable AI-based hematology diagnostics.

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It is my pleasure to express with deep sense of gratitude to Dr. Jani Anbarasi L., Associate Professor, School of Computer Science and Engineering, Vellore Institute of Technology, Chennai, for her constant guidance, continual encouragement, understanding; more than all, she taught me patience in my endeavor. My association with her is not confined to academics only, but it is a great opportunity on my part to work with an intellectual and expert in the field of Artificial Intelligence and Deep Learning.

It is with gratitude that I would like to extend my thanks to the visionary leader Dr. G. Viswanathan, our Honorable Chancellor, Mr. Sankar Viswanathan, Dr. Sekar Viswanathan, Dr. G. V. Selvam, Vice Presidents, Dr. Sandhya Pentareddy, Executive Director, Ms. Kadhambari S. Viswanathan, Assistant Vice-President, Dr. V. S. Kanchana Bhaaskaran, Vice-Chancellor, Dr. T. Thyagarajan, Pro-Vice Chancellor, VIT Chennai, and Dr. P. K. Manoharan, Additional Registrar, for providing an exceptional working environment and inspiring all of us during the tenure of the course.

We are extremely grateful to Dean Dr. Ravi Sankar A, Associate Dean (Academics) Dr. Mohana Prasad K, & Associate Dean (Research) Dr. Chitra K, of the School of Electronics Engineering, VIT Chennai, for extending the facilities of the school towards our project and for his unstinting support.

We express our thanks to our Head of the Department Dr. Vijayakumar P for his support throughout the course of this project.

We also take this opportunity to thank all the faculty of the School for their support and their wisdom imparted to us throughout the course.

We thank our parents, family, and friends for bearing with us throughout the course of our project and for the opportunity they provided us in undergoing this course in such a prestigious institution.

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Abilesh SD

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Place: Chennai

Date:

Abilesh SD

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LIST OF ACRONYMS

AI	Artificial Intelligence
ALL	Acute Lymphoblastic Leukemia
AMP	Automatic Mixed Precision
AUC	Area Under the Curve
CAD	Computer-Aided Diagnosis
CEL	Cross-Entropy Loss
CNN	Convolutional Neural Network
DL	Deep Learning
FP	False Positive
FN	False Negative
MAE	Masked Autoencoder
MSE	Mean Squared Error
ROC	Receiver Operating Characteristic
SSL	Self-Supervised Learning
TN	True Negative
TP	True Positive
ViT	Vision Transformer
XAI	Explainable Artificial Intelligence

Chapter 1

Introduction

1.1 BACKGROUND AND MOTIVATION

Leukemia is a malignant disease of the hematopoietic system, developed due to the unregulated growth of immature leukocytes in the bone marrow as well as in the peripheral circulatory system. Otherwise, the first and accurate diagnosis of this malignancy is crucial in preventing the development of the disease and also in the implementation of suitable treatment procedures. The traditional methods of detection are based on careful, manual analysis of peripheral blood smears by the use of a microscope - a time-consuming and tedious process, which requires the knowledge of trained hematologists. Inter-observer variability and possible bias are likely to occur with such subjective measures, which leads not only to the undermined reliability of diagnosis but also to a negative impact on clinical outcomes. Such intrinsic drawbacks have motivated the exploration of computational methods capable of making objective, reproducible studies of hematological images.

The fast development of artificial intelligence, specifically deep learning, has placed computer-assisted diagnosis as an interesting supplement in medical imaging. Convolutional neural networks (CNNs) have proven to be relatively successful in a range of biomedical tasks; however, they usually need large, expertly annotated datasets to be most effective. When it comes to leukemia detection, obtaining huge amounts of precisely labeled blood smear images represents a major challenge because the process of annotation requires tedious expert inspection as well as strict clinical verification. As a result, the traditional supervised learning paradigm could be ineffective in cases where there are scarce labeled data.

Recent developments in self-supervised learning and transformer-based designs have opened new avenues of learning representation in the biomedical field. Models like the Masked Autoencoder (MAE) and Vision Transformer (ViT) allow models to learn and extract strong visual representations directly on unlabeled images, eliminating the need for manual annotations. These models can be highly diagnostic when coupled with fine-tuning on relatively small labeled datasets, and they can provide interpretability based on attention-driven processes.

Bringing self-supervised transformer models to the medical image analysis, thus, presents a clear path forward to building reliable, explainable and data-efficient systems to detect leukemia. These systems have the potential to assist clinicians to provide faster and more consistent diagnoses, which will eventually improve the quality of patient care.

1.2 PROBLEM STATEMENT

Manual diagnosis of leukemia on the images of peripheral blood smear is still an elaborate and time bearing process that requires skilled hematologists. Detection of abnormal cells may be affected by the visual observer and thus discrepancy in case identification and delaying the decision to treat. Besides, the growing amount of clinical samples and the lack of trained specialists have increased the demand of the automated systems that can be used to assist pathologists during disease screening and monitoring.

Even though deep learning-based algorithms, especially convolutional neural networks, have demonstrated tremendous opportunities in the image-based diagnosis, their success heavily depends on the presence of large and properly labeled datasets. In medical imaging, such annotations are hard to come by since the annotation process should be done by an expert who has to undergo a rigorous clinical validation procedure. Consequently, models that are trained in fully supervised paradigms tend to exhibit overfitting, low generalization and low robustness to unseen data.

The other significant problem is the interpretability of deep learning models. Traditional neural networks are sometimes referred to as black boxes, which give the prediction without an obvious understanding of the morphological features that affect the decision making. Such a lack of transparency limits the clinical acceptance and integration into the diagnosis workflow.

Therefore, there is an urgent necessity of the learning framework that will be able to effectively use unlabeled data, adjust to the small number of labelled samples, and also give results that can be interpreted. To overcome these problems, the present project presents a self-supervised mnemonic ViT architecture on top of the Masked Autoencoder (MAE) preplaying to explain leukemia classification on blood smear images.

1.3 OBJECTIVES

The main objective of the undertaking is to develop and deploy an automated and interpretable deep learning architecture to classify leukemia based on microscopic blood smear images. The suggested approach will use self-supervised learning and Vision Transformer (ViT) models to reduce the use of labeled data, and at the same time improve the stature and trustworthiness of model predictions. The framework attempts to achieve clinically relevant diagnostic accuracy as well as reliable interpretability through incorporating self-supervised pretraining and supervised fine-tuning.

In the project, the following are the specific objectives:

To create self-supervised pretraining pipeline using a Masked Autoencoder (MAE) which learns intrinsic visual features on unlabeled blood smear images by reconstituting randomly masked patches, and thus enhances the level of representation.

To further optimize the pre-trained ViT encoder on labeled data based on the C-NMC Leukemia Classification dataset (Normal vs Leukemia blast), which allows it to be used successfully in binary classification between leukemic and normal cells.

In order to incorporate interpretability, e.g., attention-map visualization, and thus highlight diagnostically important cellular areas and support model focus with hematological knowledge.

To evaluate the overall performance of the models based on the measures of their accuracy, precision, recall, F1-score, and ROC-AUC to ensure that the model is worth using in generalization and diagnostic reliability in the future.

To develop a reproducible and efficient experimentation pipeline that includes optimized training protocols, validation tracking, and checkpoint management, therefore, providing reliable deployment and enabling future extensions.

By achieving these goals, the project hopes to make a contribution in advance of a data-efficient, explainable, and clinically reliable artificial intelligence model of automated leukemia detection.

1.4 SCOPE OF THE PROJECT

The current research consists of the design of a self-regulated Vision Transformer (ViT) model to perform automatic identification of leukemia with the help of microscopic images of peripheral-blood under a microscope. Image data are analyzed solely and genomic, clinical, or multimodal inputs are not used. The dataset that is used in this case is the C-NMC Leukemia Classification dataset (Gupta & Gupta, 2019) consisting of segmented images of normal and leukemia blast (malignant) cells obtained in patients. This collection of data captures real-world variations, such as small staining and light variations, thus making it appropriate in measuring the strength of the model in the context of real-world diagnostics.

In order to build a data efficient and generalizable deep-learning pipeline the pipeline is organized into two sequential steps. The first step is the self-supervised pretraining of a Masked Autoencoder (MAE) on the unlabeled part of the dataset. Random patches of images are blocked out during this stage, forcing the model to fill in the missing areas; in effect, this causes the encoder to internalise inherent morphological patterns, e.g. nuclear morphology, cytoplasmic texture and spatial organisation, without annotated labels being available.

The next step performs controlled fine-tuning, which uses the trained ViT encoder on the labeled dataset, in order to distinguish between normal and leukemia-blast cells. This operation of transferring the pretrained representations makes it faster to converge and also makes the model better at generalisation and able to pick up even subtle haematological changes even in the limited labelled data cases.

The pipeline developed focuses on three key aspects, which include efficient feature representation and stable training, as well as interpretability. The model achieved by combining ViT global attention mechanism with MAE reconstruction-based representation learning achieves strong and biologically meaningful feature extraction. The evaluation of performance is done based on quantitative metrics of Accuracy, Precision, Recall, F1-score, and ROC-AUC, which implies the adequate level of technical soundness and clinical reliability.

The ultimate deliverables include a pretrained MAE encoder, a ViT classification model which is finetuned, and the visual artefacts comprise reconstruction plot and performance plot. Together these elements form an overall, interpretable AI pipeline, and demonstrate how self-supervised ViT models can be useful towards achieving automated leukemia detection of digital blood-smear images.

1.5 MOTIVATION / SIGNIFICANCE

Diagnosis of leukemia using microscopic images of peripheral blood smear still remains a vital but complicated aspect of clinical hematology. Traditional diagnostic methods rely largely on the visual examination of red blood cells, white blood cells and platelets conducted by experienced pathologists where morphological deviations are singled out by hand to determine the subtypes of diseases. Despite decades of clinical validation and significant reliability, the method is subjective in nature and is labour-intensive; the precision of the diagnosis depends on the experience of the examiner, the quality of image resolution, and the workload that may create a discrepancy in results and delays therapeutic treatment. These gaps explain the urgency behind automated, objective, and reproducible diagnostic instruments that could assist clinicians to make timely and reliable diagnoses.

Deep-learning models, especially the convolutional neural network, have demonstrated outstanding capabilities in the image classification activity. Their use, however, is restricted by their use of large numbers of annotated datasets, which make them less applicable to the medical domain, where expert annotation is expensive and limited. This has led to the popularity of self-supervised learning wherein models can learn useful visual representations using large amounts of unlabeled data. This can be achieved through the Masked Autoencoder (MAE) paradigm which trains the models to reconstruct the distorted image patches and thus forces them to internalize the spatial and contextual structure of the images without the need to rely on manual annotations.

This method is further enhanced by the Vision Transformer (ViT) architecture that uses attention mechanisms to capture long-range dependencies and fine-grained morphological information that the medical interpretation requires. ViT models can generalize well when pretrained by self-supervision and later fine-tuned on small labeled datasets and are interpretable through visualized attention maps.

The combination of MAE and ViT is a scientific breakthrough in the explainable artificial intelligence (XAI) of hematology. The model can aid the work of medical practitioners by creating transparent attention maps, which highlight diagnostically meaningful parts, as well as encourage the trust in AI-based diagnostics.

Overall, the project is motivated by the need to fill the gap between state-of-the-art AI research and clinical application and establish a data-efficient and interpretable and resilient diagnostic framework that will help improve the validity of leukemia diagnosis in blood smear images and the overall vision of precision medicine.

1.6 ORGANIZATION OF THE REPORT

In this project report, the report is structured into six chapters, which define various phases of the research and implementation process, which begins with the contextualization of the problem and ends with the definitions of the outcomes and future improvements.

Chapter 1 - Introduction provides an overview of the problem of leukemia classification and clarifies the motivation behind the automation of the work of hematological diagnostics, and outlines the aims, scope, and relevance of the research. It also provides a schematic map of the report so as to guide the readers by the subsequent sections.

Chapter 2 - Literature Review summarizes the existing literature that is relevant to the subject of leukemia detection using traditional image-processing algorithms, convolutional neural networks, and transformer-based networks. The review anticipates the limitations of the preceding research methodologies and the research gap that this research intends to fill by using self-directed learning and explainable transformer architectures.

Chapter 3 - Methodology elaborates on the general system architecture, which includes data curation, self-supervised pre-training techniques, and Masked Autoencoders (MAE) fine-tuning on Vision Transformer (ViT), and the interpretability pipeline. The architecture, training parameters and procedural workflow are explicitly described in this section.

Chapter 4 - Implementation and Modules presents the comprehensive details of the leukemia classification system implementation. It systematically describes the stepwise execution of the pipeline, starting from dataset preparation and augmentation, through self-supervised pretraining with a Frequency-Aware Masked Autoencoder (FA-MAE) based on Vision Transformer, followed by supervised fine-tuning of a ViT classifier using pretrained encoder weights.

Chapter 5 - Results and Discussion offers the results of the empirically conducted research, the loss curves, the performance indicators, and the attention-based displays. The findings are critically analyzed as to the robustness and interpretability of models.

Chapter 6 - Conclusion and Future Work summarizes the relevant findings, contributions and possible areas of improvement or future advancement in future studies.

This hierarchical arrangement will guarantee a seamless and sequential presentation of the motivation, implementation and the empirical results of the project.

Chapter 2

Literature Review

2.1 INTRODUCTION

With the rapid development of technologies in the field of deep learning and computer vision, a significant shift in the sphere of medical image analysis has become a significant opportunity that can be used to increase the quality of the diagnostics, shorten the clinical process and improve patient outcomes. Image classification, segmentation and disease detection are examples of tasks that have been positively affected by convolutional neural networks (CNNs) and, more recently, by Vision Transformers (ViTs), which have outperformed these techniques on a variety of tasks. However, the development of such complex models usually requires large amounts of labeled data, which is still a scarce and expensive resource in medical spheres due to privacy issues, the difficulty of annotation, and the relative rarity of some conditions.

ALL is a critical field of application and automated classification in the lymphoblastic leukemia (ALL) field can provide significant clinical benefit. ALL represents about 25% of the childhood cancers, which makes it critical to diagnose the disease early to treat. Differentiating immature leukemic blast cells and normal lymphocytes at the microscopic level is not particularly easy by default due to their morphological resemblances even when the microscopic examination is performed by experienced pathologists. Deep learning-based automated diagnostic systems have the potential of solving these issues by analyzing peripheral blood smear images quickly, accurately, and reproducibly.

This literature review is a systematic analysis of the state-of-the-art methods in medical image classification, particularly leukemia detection systems. The major thematic axes of the review are the self-supervision learning paradigms, Vision Transformer, transfer-learning strategies, and dataset-specific innovations in leukemia classification. The review starts with self-supervised pre-training methods that are fundamental like Masked Autoencoders (MAEs), and continues on to the implementation of vision transformers, transfer-learning algorithms, architectural comparisons, and domain-specific applications. Specific attention is given to two-phase training methods involving unsupervised pre-training and supervised fine-tuning, a methodology that has shown exceptional effectiveness in the context of overcoming the problem of data scarcity in medical imaging. The conclusion of the review is the identification of research gaps and specification of the reason why an MAE-based Vision Transformer system is needed to address the acute lymphoblastic leukemia classification on the C-NMC dataset.

2.2 SELF-SUPERVISED LEARNING AND MASKED AUTOENCODERS

Masked Autoencoders Are Scalable Vision Learners (He et al.)

He et al.[1] presented Masked Autoencoders (MAE) as a self-supervised learning approach that conceals random patches of the input images and then reconstructs the missing pixels, and thus provides evidence of scalable learning abilities to a variety of computer vision tasks. The method uses an asymmetric encoder-decoder design, where the encoder is only fed with patches that are visible when no mask tokens are present, and a lightweight decoder is used to encode the entire image using the latent code with mask tokens. The authors noted that hiding a large fraction of the input image, of about seventy-five percent, is a nontrivial and significant self-supervisory goal. These advantages include efficient training, attainment of acceleration factors of three (or more), better predictive accuracy, scalable training of high-capacity models with robust generalization, and state-of-the-art results including eighty-seven point eight per cent accuracy on ImageNet-1K with a vanilla ViT-Huge model trained on ImageNet only. The identified limitations consist of the fact that a deliberately designed decoder is needed, even though it is lightweight; there might be semantic differences between the representations that were learned and contrastive approaches; and the critical role of decoder design in determining the quality of the learned latent representations.

Our project is based on this original study but extends it to the medical imaging field, in turn, to the classification of leukemia in the C -NMC dataset. Our similar asymmetric encoder- decoder design includes a Vision Transformer encoder and a transformer-based decoder that has the responsibility of recreating masked patches. Our approach contrasts with the natural-image pre- training He et al., by being domain specific, where the pre-training uses unlabeled cell images of leukemia. The pipeline used in the workflow consists of two steps: pre-training of MAE and subsequent fine-tuning on labeled data, fine-tuning of the masking ratio with specific consideration of medical cell images where global context is required, the acceptance of the initial pre-trained encoder weights by downstream classification, and an organized assessment of the binary classification task to distinguish between ALL blasts and normal cells.

Self-Supervised Learning for Medical Image Classification (Huang et al.)

Huang et al.[2] performed a thorough survey of self-supervised learning methods used to classify medical imagery in the 2012-2022 timeframe to summarize 79 studies to provide the researcher with implementation guidelines. The paper subdivided self-supervised approaches into contrastive learning, generative approaches, self-prediction, and hybrid approaches and evaluated their effectiveness in various medical imaging modalities. The benefits included being able to train models on large unlabeled medical datasets generating salient representations, reducing the data-annotation bottleneck that limits supervised deep learning in medical imaging, and demonstrating that self-supervised pre-training can generate performance improvements of 0.14-24.5 percent across different studies. The challenges noted included the high sensitivity of any given medical imaging task and

data to performance enhancement, the lack of standard evaluation procedures in the various self-supervised techniques, and the little research on self-supervised learning of histopathology and microscopy images compared to radiological modalities.

Our project contributes to this literature by applying generative self-supervised learning to hematology microscopy images of leukemia, in particular, by masked autoencoding, and thus providing empirical evidence to the under-researched field of self-supervised learning in hematology imaging. We contrast the performance of MAE trained on standard objects before training with that of scratch and ImageNet-init trained models, adopt the best practices defined during the review, such as the proper use of data augmentation on medical images, and use the C-NMC dataset to provide reproducible benchmarks on a public available dataset of leukemia. We also report the efficacy of self-supervised pre-training in reducing the over-fitting in fine-tuning on limited labeled data.

Self-Supervised Pre-training with Contrastive and Masked Autoencoder Methods for Medical Imaging (Wolf et al.)

Wolf et al.[3] contrasted a variety of self-supervised pre-training methods, such as contrastive learning and the Masked Autoencoder SparK (MAE -SparK), which were used in computed tomography (CT) images. The experimental procedure consisted of four different pre-training regimens that lasted approximately 30 days of training, and was followed by over 400 executions of fine-tuning on different down-stream tasks. The methods were compared using the LIDC-IDRI dataset, which includes 244, 527 CT slices of 1,010 patients and evaluated the relative performance of the methods on downstream datasets of various sizes. The results suggest that in medical imaging cases, the MAE-based approaches can be compared or even better than contrastive learning techniques. Furthermore, the findings indicate that the advantages of self-supervised pre-training depend on the magnitude of the downstream dataset and the greatest benefits are experienced in the case of a limited data set. Notably, the paper also presents evidence that domain self-supervised pre-training is better than ImageNet pre-training on medical tasks. The first weakness of the work is the high computational costs of the pre-training step, each regimen took about 30 days on high-performance graphics processing units, and secondly, it focused exclusively on CT imaging, which might be problematic in generalizing to other modalities, like microscopy or histopathology. Also, the analysis was limited to a small scope of the downstream activities, thus potentially limiting the depth of the analysis of the quality of representation.

Our project can be seen as applying the lessons of this cross-comparative study, with the pre-training based on MAE specifically applied to microscopy images of blood cells, thus covering a much more different modality than CT scans. We perform a systematic analysis of the pre-training effectiveness in the circumstances of data sparsity that are relevant to leukemia datasets. The study uses training strategies that are computationally efficient, such as optimised batch-size selection and learning-rate scheduling. We compare MAE pre-trained models to baseline methods that include random initialisation and the fully supervised training.

Lastly, we also record dynamics of training, including the loss convergence curves and evidence of overfitting that is witnessed during the pre-training as well as during the fine-tuning stages.

Self-Supervised Multiple Instance Learning for Acute Myeloid Leukemia Classification (Kazeminia et al.)

Kazeminia et al.[4] examined self-supervised learning (SSL) as a pretraining method of Multiple Instance Learning-based subtype classification of Acute Myeloid Leukemia (AML) in peripheral blood smears. The authors investigated three of the most prominent paradigms of SSL: SimCLR, SwAV, and DINO, and compared the performances of these models with those of a supervised baseline in pretraining. The findings showed that the performance of the encoders trained using SSL matched that of the encoders trained using supervised data and the labeled data were not required to train the encoders. This observation indicates an economical and data efficient method of AI-based diagnostic procedures, especially in settings where annotations are limited or costly, and validates the relevance of SSL to hematological imaging modalities in which cell-level labeling is challenging. However, the study was limited to AML, which did not allow direct comparisons with Acute Lymphoblastic Leukemia (ALL) specific datasets; the experiments also used fairly small data splits, which may conceal the scalability of SSL methods, and were also limited to contrastive methods without generative SSL variations like masked autoencoders.

We will continue this line of questioning by focusing on ALL instead of AML, using the publicly available C-NMC dataset and using a generativeSSL framework (masked autoencoders) instead of contrastive learning. We also solve a binary classification problem (ALL vs normal) versus the multiclass subtype context of the previous study, and we present an end-to-end pipeline consisting of self-supervised pretraining, supervised fine-tuning, and strict performance testing.

2.3 VISION TRANSFORMERS FOR MEDICAL IMAGE CLASSIFICATION

An Improved Vision Transformer for Medical Image Classification (Halder et al.)

Halder et al.[5] explored the usability of ViT models in biomedical image classification by using ViT-base-patch16-224 model pre-trained on ImageNet-21k and later fine-tuning on four subsets of MedMNIST, namely BloodMNIST, BreastMNIST, PathMNIST, and RetinaMNIST. The performance measures achieved were above the previous performance measures with the accuracy of 97.90, 90.38, 94.62 and 57 respectively. The paper has highlighted a number of strengths of ViT: the model has an inherent self-attention mechanism that allows modeling of long-range dependencies otherwise unobservable by conventional convolutional neural networks (CNNs) since their receptive fields are inherently local; the model has shown to be able to make significant accuracy improvements when applied to blood-cell classification tasks that are directly relevant to leukemia diagnosis; and the model has shown to be able to support a variety of medical imaging modalities through appropriate fine-tuning. However, the authors also noted a number of weaknesses: the computational cost of training and inference is significantly greater than with lightweight CNN architectures; the use of such large-scale generic pre-training datasets as ImageNet-21k leads to possible domain shift when used in medical tasks; and hybrid architectures that act synergistically by combining CNN and Transformer elements have not been fully explored.

On the basis of this work, we came up with a ViT-based model to classify acute lymphoblastic leukemia (ALL) on the C-NMC dataset which has a significantly larger and more difficult corpus than BloodMNIST. We use the technique of Masked Autoencoder (MAE)-based self-training on domain-specific leukemia images, which does not necessarily use ImageNet as the initial point of pre-training. Two-stage training is used, where the model is first trained to learn representations of leukemia-specific features, and then fine-tuned to classify; it is ensured that the entire ViT encoder is retained by transferring weights carefully to preserve its learned representations. Extensive evaluation of performance is conducted, with accuracy, F1-score, confusion matrix, and receiver operating characteristic (ROC) being some of the metrics.

Vision Transformers in Medical Imaging: A Comprehensive Review (Aburass et al.)

Aburass et al.[6] introduced a detailed survey of Vision Transformer applications in various fields of medicine imaging such as breast cancer, brain tumors, lung diseases, and blood disorders critically reviewing architectures, transfer learning methods, and understanding of the models. The review summarized the results indicating that ViT usually performs better than conventional CNNs at the medical imaging classification. Among the advantages were offering evidence of the better performance of ViT in the various fields of medical imaging such as hematology, outlining the critical challenges such as data scarcity, model interpretability, and computational costs to achieve optimal performance by ViT in medical tasks. Limitations were a dearth of standardized evaluation procedures across various

studies that makes it hard to compare performance across different studies, little discussion of computational efficiency and deployment issues in clinical settings, and inadequate coverage of hybrid architectures which could potentially be advantageous combinations of CNNs and Transformers.

To fill in various gaps found in this review, our project proposes a complete ViT-based system with domain-specific pre-training to classify leukemia, providing consistent evaluation metrics on the publicly available C-NMC dataset, describing computational requirements and training dynamics to enable reproducibility, setting up efficient training schemes, such as suitable schedule of learning rate and data augmentation, and providing empirical evidence of ViT effectiveness itself in blood cell classification as a microscopy sample.

MedViT: A Robust Vision Transformer for Generalized Medical Image Classification (Manzari et al.)

Manzari et al.[7] introduced MedViT which is a hybrid model that integrates convolution blocks with transformer blocks to provide strong performance in medical image classification. It is designed to take advantage of both local feature extraction of convolutions as well as global context modeling of transformers. Among its advantages were the ability to attain state-of-the-art performance on a variety of medical imaging tasks with effective CNN and Transformer fusion, the ability to be more resistant to domain shift than pure CNN or Transformer architectures, and an architecture that can be generalized across multiple medical imaging domains without requiring a large quantity of task-specific adjustments. Drawbacks consisted of more architecture complexity and the number of parameters than standalone architectures, the need of very careful hyperparameter tuning to achieve a balanced CNN and Transformer, and more computational demands in both training and inference than lightweight models..

Our project follows a different architecture by using a pure Vision Transformer encoder than a hybrid CNN-Transformer architecture, the rationale being that MAE pre-training can deliver the inductive biases that the convolutional layers provide. Specifically, we are interested in leukemia classification instead of building an overall medical imaging architecture, self-supervised pre-training to acquire domain-specific features, which decrease the complexity of the architecture, and the direct comparison between the MAE-ViT system and more traditional supervised training to evaluate the performance of this simplified architecture with suitable pre-training.

2.4 MASKED AUTOENCODERS AND ASYMMETRIC ENCODER-DECODER ARCHITECTURES

Asymmetric Decoder Design for Efficient Convolutional Encoder-Decoder Architectures in Medical Image Reconstruction (Rahman et al.)

Rahman et al.[8] applied to the medical image reconstruction problem the asymmetrical encoder-decoder architectures, in particular, to undersampled MRI reconstruction with the Calgary Campinas dataset. The proposed architecture was made up of EfficientNet as an encoder, which was combined with a custom decoder that combined multi-resolution features by incrementally adding features. Among these advantages were showing that asymmetric architectures that use powerful decoders that concentrate on high-frequency features have higher pixel-level accuracy, four times less inference time than conventional U-Net with no or better quality, and that asymmetric architectures are capable of much higher pixel-level accuracy than symmetric U-Net architectures. Limitations were that it focused on reconstruction tasks as opposed to classification that makes direct applicability to diagnostic tasks, evaluation was restricted to MRI data that is different in nature to microscopy images, and the architecture may be difficult to implement on other medical imaging modalities.

We use the concept of asymmetric encoder-decoder design but in self-supervised learning as opposed to direct reconstruction. We use an effective ViT encoder to extract features and a more lightweight decoder to do masked patch reconstruction but use the learned representations of the encoder in downstream classification and design the decoder to be powerful enough to pre-train and at the same time, be computationally efficient.

A Survey on Masked Autoencoders for Visual Self-Supervised Learning (Zhang et al.)

A detailed review of masked autoencoder-based visual self-supervised learning methods was given by Zhang et al.[9] comparing MAE and SimMIM with many subsequent studies. The analysis of the relationship between masked autoencoders and joint-embedding techniques such as contrastive learning was performed in the survey. Among the benefits, it was found that the masked autoencoders perform worse at the linear probing accuracy than contrastive methods, and that it can be explained by the fact that MAE is designed to focus on low-level patch matching, whereas contrastive learning is designed to capture high-level semantic features, and that combining the two objectives can be beneficial. Some of the constraints were the rapid advancement of the field which made it difficult to cover the field comprehensively, lack of empirical comparison of various medical imaging fields, and lack of guidelines on how to apply various approaches of the SSL to certain applications

This literature body is the contribution of our project that empirically shows that MAE is effective particularly when applied to microscopy-based medical imaging. We test pure MAE without contrastive components to determine its standalone performance, test both the fine-tuning and the linear probing learning, and

interpret the results of these tests on the quality of representations, trade-offs between the pre-training computational cost and downstream performance improvements, and give specific implementation details as a guide to researchers who may work with similar medical imaging datasets.

2.5 TRANSFER LEARNING AND FINE-TUNING STRATEGIES

Comparison of Fine-tuning Strategies for Transfer Learning in Medical Image Classification (Davila et al.)

Davila et al.[10] is an extensive study of different fine-tuning techniques used on pre-trained models of different medical imaging domains such as X-ray, MRI, Histology, Dermoscopy, and Endoscopy. Eight strategies were tested such as full fine-tuning, linear probing, gradual unfreezing, and adaptive learning rates on three CNNs (ResNet-50, DenseNet-121, VGG-19). Advantages in the form of proving that linear probing with full fine-tuning could yield significant improvements in more than half the cases tested, proving that adaptive learning rate methods such as Auto-RGN could result in a performance improvement up to 11 percent on specific modalities, and finding that DenseNet architecture performed better on alternative fine-tuning methods than on traditional full fine-tuning. The constraints were: CNN architectures were only evaluated and no attempt was made to investigate strategies to fine-tune a Vision Transformer, only supervised transfer learning on ImageNet without understanding self-supervised pre-training and only limited exploration of strategies under conditions of extreme data scarcity (as seen in rare diseases).

We build on this work by analyzing fine-tuning techniques on Vision Transformer models, but those pre-trained on MAE. Our use of full fine-tuning instead of layer-wise unfreezing relates to the relative small size of the C-NMC dataset, the adaptive schedules of learning rates implemented using the ReduceLROnPlateau framework, the comparison between MAE pre-trained models and ImageNet baselines and random initialization baselines, and the reporting of the interaction between self-supervised pre-training and supervised fine-tuning strategies.

Transfer Learning Pre-training Dataset and Fine-tuning Effect Analysis in Cancer Histopathology Images (Howlader et al.)

Howlader et al.[11] conducted the analysis of transfer learning pre-training dataset and fine-tuning effect on cancer histopathology imaging by testing the three deep neural network algorithms on three target datasets in different fine-tuning setups. The experiment was a comparison between pre-training on ImageNet and histopathology-specific data. These advantages were the ability to show that performance increases with more images utilized during fine-tuning, which was not prioritized in most studies, and that pre-training on cancer histopathology datasets does not always perform better than ImageNet pre-training or random initialization, and to show that task-specific benefits of domain-specific pre-training are task-dependent. The weaknesses were a focus on histopathology as opposed to other medical imaging modalities, an examination of CNN architectures without

considering Transformer-based architectures, and an examination of self-supervised pre-training methods beyond supervised ImageNet transfer.

The limitations of our project are that it involves self-supervised MAE pre-training on domain-specific leukemia cell images, measures the performance of this method against ImageNet initialization, and shows evidence of the importance of domain-specific self-supervised pre-training to microscopy-based medical imaging.

Transfer Learning in Medical Image Segmentation (Karimi et al.)

Karimi et al.[12] gave a critical evaluation of transfer learning in training fully convolutional networks to medical image segmentation to examine the effect of transfer learning on model parameters and learned representations. It was found that convolutional filters do not vary much and that accurate models can be trained by freezing the encoder at random values and training the decoder only. Among the benefits were the ability to provide evidence to prove the high task and data-dependence of improvements in transfer learning where large gains are only observed when segmentation tasks become more challenging, the fact that feature reuse is not just confined to the first few layers but can be far more substantial in later layers, and the variability of models trained through transfer learning can be as high as that of models trained through randomization. Limitations were focus on segmentation tasks and not classification, evaluation was mostly based on CNNs though not Vision Transformers, and little empirical investigation was done regarding the alternative of self-supervised pre-training as an alternative to supervised ImageNet transfer.

The findings are complemented by our project because our approach is not segmentation but classification, self-supervised pre-training with a specific domain focus, implementation of the Vision Transformer architecture in which transfer learning dynamics can be different than CNNs, and analyses of learned representations via fine-tuning performance and overfitting behavior.

2.6 LEUKEMIA CLASSIFICATION DATASETS AND METHODS

C-NMC 2019: ALL Challenge Dataset of ISBI 2019 (Gupta et al.)[13]

C-NMC dataset C-NMC dataset was published at the ALL Challenge at ISBI 2019, containing over 15000 carefully curated high-resolution cancer cell images. The repository contains a training set of 10 661 images (7 272 images ALL-positive and 3 389 normal cell images were obtained of 73 subjects), a pre-test set of 1 867 images, and a final test set of 2 586 images. The main strengths of the release are the availability of a curated resource on B-lineage acute lymphoblastic leukemia of the largest publicly available size, thus allowing benchmarking to become standardized; the quality of annotations that are carried out by trained hematologists, thus reducing the effects of label noise; and the ability to reproduce research due to the open availability of information that eliminates an essential gap in medical artificial intelligence. However, the dataset has a number of limitations: it has a sharp class imbalance, namely, 68-percent cancer and 32-percent normal cells, which can be problematic in terms of bias when being used in training the model; the images are single-cells instead of whole-slide, so it is not as applicable to clinical practice.

The C-NMC dataset is the main source of benchmark of our MAE-ViT methodology in our project. In order to reduce the natural imbalance of classes, we use advanced data augmentation policies and warped loss functions on fine-tuning. Our training set is 6 033 images in three strata, and we use the complete set as a pre-training set, and we use careful train-validation partitioning which ensures patient-level separation, thus avoiding data leakage. Lastly, we perform a complete analysis on the validation sample of 1867 images in order to critically test the overall performance of the model with regard to generalization.

Hybrid Techniques for the Diagnosis of Acute Lymphoblastic Leukemia (Ahmed et al.)

The article by Ahmed et al.[14] explored a two-stage deep learning method in the diagnosis of acute lymphoblastic leukemia (ALL) based on C-NMC 2019 and ALL-IDB2. The methodology involved first performing active contour segmentation of leukocyte images, and subsequently, extracting features using convolutional neural network (CNN) and performing ensemble classification. The feature extraction was done using the DenseNet121, ResNet50 and MobileNet architecture and the classification phase was done using random forest (RF) and XGBoost algorithms. The hybrid system produced an accuracy of 98.8 per cent on C-NMC dataset and a 100 per cent on ALL-IDB2, thus showing the effectiveness of CNN-machine-learning ensembles in leukemia classification. Another point brought up in the study was the possibility of improving the performance of several CNN models with feature fusion to overcome the performance of individual models. Nonetheless, the given methodology was flawed by a number of shortcomings: the multi-stage pipeline based on the need to use a complicated white-blood-cell segmentation algorithm was inconveniencing in terms of implementation complexity and computation costs; the requirement to have an accurate white-blood-cell segmentation algorithm was also an

issue in case of poor image quality; and the platform was not attempting to seek out end-to-end deep learning techniques, which could simplify the processing pipeline.

The present project compares with Ahmed et al. since it uses an end-to-end Vision Transformer (ViT) classification pipeline that does not have the explicit segmentation phase. ViT model is pre-trained on a masked auto-Encoder (MAE) task that allows learning strong features representations using the un-segmented whole-image inputs without reference to linked segments. The architecture eliminates the high computational overhead that would otherwise be incurred with an ensemble strategy by using only one model, without reducing diagnostic performance. The overall workflow, including pre-training and fine-tuning, is also simplified to facilitate further applications to clinical practice.

Recognition of Acute Lymphoblastic Leukemia Cells in Microscopic Images (Amin et al.)

Amin et al.[15] proposed a complex system of recognizing and subtyping the acute lymphoblastic leukemia (ALL) based on blood and bone-marrow smears microscopy. They have used a k-means clustering technique to segment the nucleus, extract geometric and statistical features and support-vector-machine (SVM) classification that were tested using ten-fold cross-validation. The final accuracy of the performance was 97.0000 cent in the detection of malignant cells and 95.6000 cent in differentiating the subtypes L1, L2, and L3 and separating the normal lymphocytes correctly. Its method is interpretable, based on morphology features that can be consistent with the accepted clinical diagnostic criteria, and even attains strong performance with the standard machine-learning methods, and is multi-class differentiable without huge deep-learning resources. Some of the weaknesses of the approach are that it uses hand-crafted features requiring domain knowledge and can fail to capture subtle patterns, it requires manual thresholding to achieve segmentation, and it is not as scalable as end-to-end deep-learning systems.

This paper aims to take a paradigm shift toward end-to-end deep-learning methods that are able to learn discriminative representations directly on raw images. We use Vision Transformers to directly learn representations, using self-supervised pre-training to discover useful features without specifying features of interest. It centers on binary classification, i.e. the separation of ALL and normal, which offers a clinically relevant screening device, and we use a fully automated pipeline including image-getting, pre-processing, feature-extraction and the final classification.

2.7 DATA AUGMENTATION AND OPTIMIZATION IN MEDICAL IMAGING

A Deep Learning Image Data Augmentation Method for Single Tumor Segmentation (Zhang et al.)

Téassse et al.[16] suggested a new data augmentation process of tumour segmentation that forms full tumour images that are similar to the regional tumour images produced by the horizontal, vertical, and diagonal cutting and mirroring of augmentation. In conjunction with Mask-RCNN and U-Net, the approach attained an improvement of Dice coefficient (DSC) of 9.66 -13.74 (percentages) compared to the non-augmented baseline on breast ultrasound data. It had the following benefits: it generated new samples using real images whilst maintaining the true tumour properties, offered a better sample independence than the traditional augmentation that changes pixel neighbour relations and generated tumour samples that have well-defined boundary information that is friendly to precise segmentation. The limitations included its design that was specific to single tumour segmentation and thus limited to classification problems, a need to ensure accurate localisation of the tumour centre that might not be easily achieved in practice and an experiment that was limited to breast ultrasound data with no data available on other imaging modalities.

Our project uses standard augmentation methods that are appropriate in cell-level classification, instead of using complex methods of cutting augmentation. We use random horizontal flipping and rotation to make training data more diverse, use small amounts of augmentation in masked auto-encoder pre-training to maintain natural image statistics, use stronger amounts of augmentation in supervised fine-tuning to reduce overfitting, and scrupulously record the effects of augmentation methods on model performance and generalization.

Comparing Different Deep Learning Architectures for Classification of Chest Radiographs (Bressem et al.)

Bressem et al.[17] conducted a thorough comparative study of fifteen neural network models (VGG -16, ResNet -50, and DenseNet -161) on the CheXpert data set of chest X-rays classification. It tested the area under the receiver operating characteristic curve (AUROC) as well as the training time in a variety of pathologies. Their results showed that ResNet and DenseNet models were always highly performing in relation to a large variety of pathologies; they also determined that VGG-16 is capable of offering the most practical trade-off between the two metrics when both are picked together (AUROC and area under the precision-recall curve). Notably, the paper proposed that the enhancement of the network depth is not always accompanied by the high-performance, which means that the optimal architectural depth depends on the task. However, the study had limitations such as being narrowed only to the classification of chest X-rays, no consideration of microscopy or other imaging modalities, no coverage of Vision Transformers or hybrid model architectures, and minimal examination of self-supervised pre-training other than supervised ImageNet transfer.

We apply these architectural understanding in the current project by incorporating a Vision Transformer model instead of the traditional convolutional neural networks. We use masked autoencoder (MAE) self-supervised pre-training that is customized to our target domain of microscopy of blood cells, which have significantly different imaging properties as compared to radiographs. Moreover, we offer a comprehensive performance evaluation including accuracy, F1 -score, and confusion matrices.

Adam Optimizer and Learning Rate Scheduling (Kingma et al.)

Kingma et al.[18] also presented Adam (Adaptive Moment Estimation), a stochastic gradient descent optimizer which dynamically scales gradient learning rates on a per-parameter basis using exponential moving averages of gradients and squared gradients. The algorithm is computationally efficient, has low memory consumption, diagonal rescaling of gradient invariance, and is optimally adapted to high-dimensional parameter space and massive data sets, and converges faster than standard SGD with tuned hyperparameters. However, Adam can have a tendency to converge to suboptimal solutions, is sensitive to the selection of hyperparameters, especially the learning base learning rate and decay coefficients, and can diverge when an appropriate learning-rate schedule is not used, especially on later training runs.

Adam is the main optimiser in the current research. A starting learning rate of 3×10^{-5} is chosen due to empirical validation. Learning rate is then adjusted by a ReduceLROnPlateau scheduler which reduces the rate when validation performance levels off. The weight decay (L2 regularisation) is used to alleviate overfitting. These training dynamics are recorded during the process of optimisation and they consist of loss trajectories and learning-rate adjustments.

2.8 CHALLENGES AND RESEARCH GAPS IDENTIFIED

Even though a lot of progress has been made in the field of self-supervised learning, Vision Transformers, medical image classification, there still are a number of challenges and gaps in the research in the context of leukemia classification and automated diagnostics in hematology. A lack of data and their quality is the main issue, as it is proven by research on C-NMC dataset (Gupta et al.; Ahmed et al.), where the number of annotated samples and the lack of class balance complicate building models, especially when rare types of leukemia are to be considered. Domain-specific pre-training has not been studied extensively, and most researchers use ImageNet initialisation (Halder et al.; Manzari et al.), causing domain shift, whereas self-supervised pre-training on medical images in the context of microscopy in particular has been underrepresented relative to radiological modalities (Huang et al.; Wolf et al.).

Another important gap is in architectural optimisation, where most leukemia classification systems use CNN-based methods (Ahmed et al.; Amin et al.) but Vision Transformers have shown potential in other medical imaging tasks (Aburass et al.), and masked autoencoder approaches to hematology imaging have not been fully investigated (Kazeminia et al.).

Vision Transformer fine-tuning medical imaging Fine-tuning Vision Transformers are not adequately researched, and much of the literature on CNN fine-tuning has been done (Davila et al.; Howlader et al.), without providing specific recommendations on how to achieve the best fine-tuning in ViTs on small medical datasets.

Computational efficiency and pragmatic deployment factors are often disregarded and studies have demonstrated high accuracy, but have used a complex multi-stage pipeline (Ahmed et al.) or ensemble methods that are computationally infeasible in real-time clinical applications. Furthermore, the records of the training times, resource necessities and inference rates needed to be deployed to the clinical settings are frequently inadequate.

There is no standardisation of evaluation cross-studies have different split of datasets, evaluation metrics, and cross-validation methods, as a result, evaluation is not as easily compared as required, and no progress is made by lack of standardisation of evaluation specifically to leukemia classification.

The specified gaps prompt the creation of the proposed methodology, which handles the issue of data scarcity by self-supervised masked autoencoder pre-training on unlabeled leukemia images, a Vision Transformer architecture that is suggested to be optimised by cell-level microscopy classification, detailed documentation of the training settings, including learning rates, schedulers, and using augmentations, and an end-to-end pipeline featuring pre-training and fine-tuning on leukemia images to be deployed in practice. Also, it sets reproducible assessment protocols on the standard C-NMC benchmark data.

2.9 SUMMARY

The literature review was a systematic review of the current methodologies of medical image classification, which focused on leukemia detection systems and also focused on self-supervised learning, Vision Transformers, transfer-learning methods, and application in specific domains.

Preliminary experiments with Masked Autoencoders (He et al.; Zhang et al.) showed that generative self-supervised learning was effective in computer vision, and later experiments (Wolf et al.; Huang et al.) on medical imaging showed that it could help overcome data-scarcity issues.

Vision Transformer models have become such potent replacements of convolutional neural networks in medical imaging (Halder et al.; Aburass et al.; Manzari et al.), with its self-attention mechanisms allowing it to better model long-range relationships that are essential to detect subtle morphological changes between diseased and healthy cells.

Transfer-learning and fine-tuning strategies (Davila et al.; Howlader et al.; Karimi et al.) demonstrate that the optimal strategy is highly task and architecture specific, and that in medical imaging tasks, self-supervised pre-training on domain specific data can be better than ImageNet pre-training.

Standardized benchmarking of ALL classification has been made possible by the availability of the C-NMC dataset (Gupta et al.), and the recent methods have demonstrated impressive performance using hybrid CNN-machine learning systems (Ahmed et al.; Amin et al.). However, they often make use of complicated multi-stage pipelines or require clear segmentation and manual features.

Techniques of data augmentation (Zhang et al.; Bressemer et al.) and optimization methods, such as the Adam optimizer and learning-rate scheduling, have been found to be important to train strong models given small medical imaging data. The choice of augmentation, the choice of optimizers, and the choice of architecture have a large effect on the performance of the model, and researchers have shown that careful hyper-parameter tuning can give significant gains.

To overcome the discovered gaps in research, the proposed work introduces a system based on a Masked Autoencoder and a Vision Transformer, which is specifically trained to classify acute lymphoblastic leukemia. The two-step method is a hybrid of achieving domain-specific representations using self-supervised pre-training on unlabeled C-NMC images, and binary classification by performing supervised fine-tuning on labeled data. This structure offers an end-to-end platform that does not require the use of explicit segmentation or feature engineering, uses the representational power of Vision Transformers that are trained via domain-specific self-supervision learning, solves the issue of data scarcity via effective use of unlabeled images during pre-training, and provides reproducible evaluation protocols with full performance analysis comprising of accuracy, F1-score, confusion matrices, and ROC curves. This practice will be beneficial to the area of AI-based hematology diagnostics by filling the methodological and empirical gaps between advanced self-supervised learning methods and real-world medical image classification.

Chapter 3

Methodology

3.1 INTRODUCTION

In the chapter, the solid and data-efficient deep-learning structure used to classify leukemia in peripheral blood smear pictures is outlined in details. The suggested architecture uses a two-step design, combining self-supervised pretraining with supervised fine-tuning, and therefore, enables the effective utilization of unlabeled and labeled image data.

At the first stage, a Masked Autoencoder (MAE) based on the Vision Transformer (ViT) architecture is trained on a large amount of unlabeled blood-smear images obtained in the C-NMC Leukemia Classification dataset. The MAE learns to encode inherent morphological features, such as nuclei, cytoplasm, and staining variations, without human intervention by conducting stochastic masking of image patches and following it up with reconstruction.

The second stage involves fine-tuning on the labeled subset of the dataset with a supervised ViT classifier which has the pretrained encoder embedded within it to classify the binary discrimination between the normal and Leukemia blast cells. This plan makes use of self-monitored representations to enhance high convergence and generalization when there is limited annotated data.

All of the implementation was done with PyTorch combined with the Hugging Face Transformers library in Google Colab which was used to ensure computational efficiency and reproducibility. The retrospected approach is a means to provide an interpretable and clinically reliable leukemia-classification system.

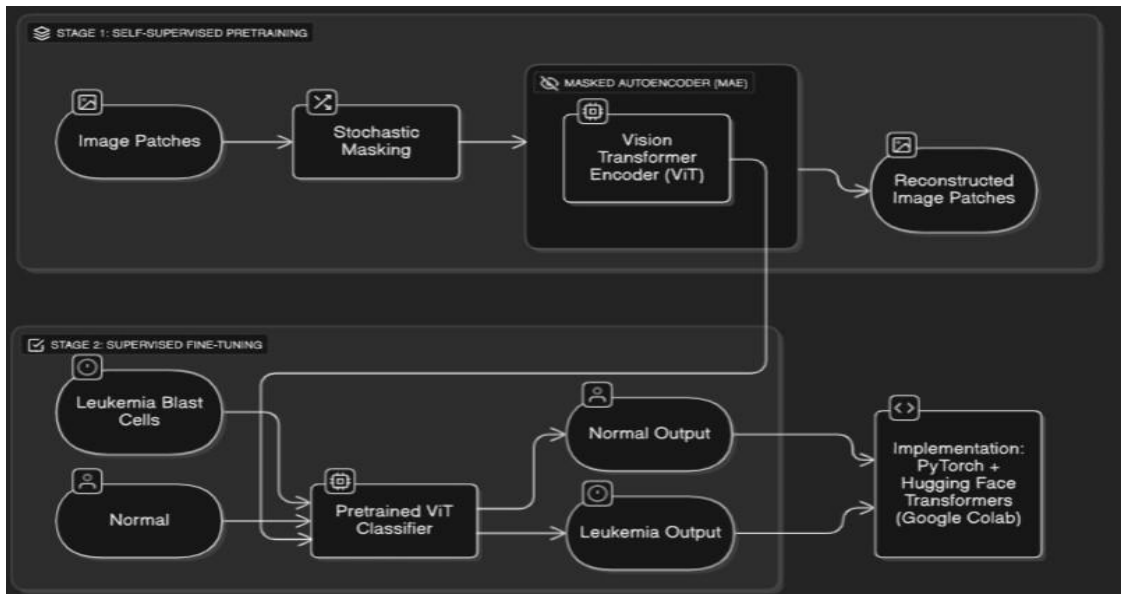


Figure 3.1: Proposed Two-Stage Architectural Workflow

3.2 DATASET DESCRIPTION AND PREPARATION

This chapter outlines the overall methodology used in building a robust and data efficient deep learning model in automatically classifying leukemia using peripheral blood smear images. The architecture that will be proposed entails a two-stage architecture that combines self-supervised pretraining and supervised fine-tuning, thus enabling the effective use of unlabeled and labeled image data.

The first stage involves a Masked Autoencoder (MAE), which is based on the Vision Transformer (ViT) model, trained on a large dataset of unlabeled images of blood-smear found in the C-NMC Leukemia Classification collection. Through patchwise masking and masking it back, the MAE is taught to encode intrinsic morphological features, including nuclei, cytoplasm and staining variation, without using any manual labels.

During the second stage, the trained encoder is then integrated into a supervised ViT classifier, and it is fine-tuned on the labeled subset of the data to oversee binary classification of the Normal and Leukemia blast cells. The approach uses self-supervised representations to achieve better convergence and generalization with small annotated data.

All computations and implementation were done in PyTorch and the Hugging Face Transformers library on Google Colab, thus making it computationally efficient and reproducible. The systematic approach results in the interpretation and clinically dependable leukemia-classification system.

Category	Subtype(s)	Usage in Project	Purpose
NORMAL (ALL)	Normal lymphocytes (healthy lymphocytes)	Used in both MAE pretraining (as unlabeled data) and fine-tuning (as class 0)	Represents healthy cells
Leukemia (HEM)	Leukemia blast cells (abnormal immature WBCs)	Used in both MAE pretraining (as unlabeled data) and fine-tuning (as class 1)	Represents malignant cells
Dataset Split	Train / Validation (custom split)	Used for supervised training and evaluation	Ensures balanced and reproducible validation

Table 3.1: Dataset Description

3.3 SELF-SUPERVISED PRETRAINING USING MASKED AUTOENCODER (MAE)

The pretraining of the first phase in the proposed pipeline is self-supervised and based on the Masked Autoencoder (MAE) built on a Vision Transformer (ViT-Base) backbone. The learner of visual representations of white blood cells with no explicit labels gains an inherent concept of the cellular morphology, thus, enabling it to acquire discriminative patterns on the next, and ultimately, supervised fine-tuning stage. Simply put, the MAE creates structural patterns of cells before they are subjected to their class labels, i.e.: Normal or Leukemia blast.

The MAE has two main parts which include encoder and decoder. The encoder, inspired by the ViT -Base architecture, divides every input image into 16 -non-overlapping patches with 16 pixels. These patches are flattened and mapped into feature vectors, and then in combination with positional encodings, to maintain the spatial relationships. The sequence is then passed through a series of transformer layers using multi-head self-attention and feed forward sub-layers that help the encoder to pick local cellular detail as well as the global context.

The decoder is a lightweight transformer which re-assembles the original image with partial information. It is presented with the coded visible patches along with masks that mark the areas covered and it is trained to reconstruct the missing information. Since the encoder can only see a portion of the image it has to guess at what the encoder can not see so that it learns meaningful and generalisable features and not memorises superficial textures. Such an asymmetric construction, a high powered encoder and a light-weight decoder, gives this computational efficiency with high representational capacity.

Random masking of image patches is done in around 75 percent of the iterations and the masked pixel patches are substituted with zero tokens which eliminate pixel information. This stochastic masking technique will guarantee the model an extreme number of combinations of visible patches and hidden patches, and increase its ability to generalise to the many different appearances presented by blood-smears. The loss is the Mean Squared error (MSE) of the reconstructed and original images but only at masked patches. This facilitates proper reconstruction of invisible areas hence facilitating easy optimisation and stable learning.

Known as AdamW optimiser, training is performed with the aid of a cosine learning-rate schedule to achieve gradual convergence. Automatic Mixed Precision (AMP) is also used to improve faster computation and lessen the use of memory on the GPU at the expense of accuracy. The loss of reconstruction is followed through epochs to check the progress of learning.

After pretraining, the encoder generates visually consistent reconstructions of masked blood-smear images, which suggests that it has learned nuclear, cytoplasmic, and staining patterns of hematological structures. The encoder parameters are only saved under `mae_encoder.pth` because only during reconstruction, the decoder is needed.

The pretraining phase allows the model to capture the morphological diversity of thousands of unlabeled blood-smear images sampled by the C-NMC Leukemia Classification dataset, which includes variations in nuclear geometry, cytoplasmic density and staining intensity. Through the acquisition of these large visual features that are not marked in any way, the MAE provides the ViT with a powerful initialization that captures the visual language of blood cells.

To conclude, the MAE pretraining step provides the Vision Transformer with strong contextualization of hematological images before any supervised learning is performed. Through the reconstruction of masked image patches, the model learns both spatial and textural dependencies which are then fine tuned to be used in classification, which forms the backbone of the proposed leukemia-classification framework.

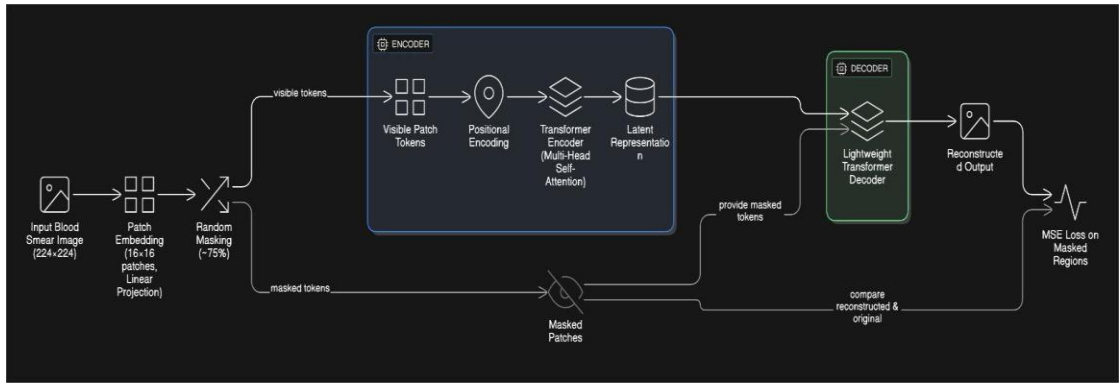


Figure 3.2: Masked Autoencoder (MAE) Pretraining Architecture

MAE Encoder Architecture

The core of the Masked Autoencoder (MAE) is its encoder, which is responsible for learning the latent representations from the visible patches. For this project, a standard Vision Transformer 'Base' (ViT-Base) architecture was deliberately chosen. This decision was based on its proven balance between high model capacity and computational trainability.

The specific hyperparameters of this encoder are detailed in Table 3.2.

Hyperparameter	Value	Description
Patch Size	16x16	The input image (224×224) is divided into 196 patches of this size.
Number of Layers	12	The total number of stacked Transformer blocks in the encoder.
Embedding Dimension	768	The size of the vector representing each image patch (and the 'hidden size' within the Transformer).
Number of Heads	12	The number of parallel attention heads in each multi-head self-attention layer.
Total Parameters	~86 Million	The approximate number of trainable parameters in the standard ViT-Base model.

Table 3.2: ViT-Base Encoder Hyperparameters

3.4 SUPERVISED FINE-TUNING OF PRETRAINED ViT ENCODER

The current study after the self-supervised pre-training of the MAE, passes to the supervised fine-tuning phase, which involves the training of the pretrained Vision Transformer (ViT) encoder to be highly accurate in distinguishing between normative and leukemic blast blood-smear images. The phase uses the annotated subset of the C-NMC Leukemia Classification data, which consists of two classes of the healthy and malignant white-blood-cell morphologies. The advantages of starting with MAE-derived weights as opposed to random parameters are that the Vision Transformer accelerates convergence, attains greater stability and higher generalization in the conditions of having a small number of labeled samples and being exposed to morphological structures that are inherent in the unlabeled corpus.

The ViTForImageClassification model, which is a part of the Hugging Face Transformer library, is used to perform the fine-tuning with the number of labels set to 2 so that binary classification could be performed. The model is initialized with ImageNet-pretrained weights, which gives it a generic visual prior, and then partially overwritten with encoder parameters ('mae_encoder.pth') which are the result of the MAE pre-training phase. This architecture has been successfully used to transfer self-supervised representations to the downstream classification model so that the ViT backbone is allowed to maintain low-level textual acuity and high-level hematological awareness.

Each input imagery is resized to 224 224 pixels and brought down to the [0, 1] range. In order to be more robust and reduce overfitting, data augmentation is used in training; specifically, random horizontal flips, small rotations (± 10 -15 degrees), and colour jittering are used to simulate microscopic and staining variability. The data will be split randomly into training and validation subsets, and the proportions of classes within the subsets will be equal. PyTorch DataLoader instances are used to load batches efficiently, hence allowing computation by the use of a GPU.

The Cross Entropy Loss (CEL) is used to estimate the distance between the estimated probabilities and actual class labels in model optimisation procedure. The AdamW optimiser (Learning rate = 3×10^{-5}) is used to update the parameters, and the regularisation by the weight-decay is also added to encourage convergence stability. ReduceLROnPlateau scheduler is used to track the validation loss and decrease the learning rate when the progress of the learning process is not good, providing a gradual reduction of the learning rate to the optimal performance. The training lasts about twenty epochs, model checkpoints (vit_finetuned_mae.pth) are maintained at a time when there is an improvement in validation accuracy.

The training and validation measures (loss, accuracy and F1-score) are logged and their curves against the epochs are plotted to visualise the convergence dynamics. The resulting curves help in showing that pre-training helps the model to adapt to the classification task faster resulting in upwardly smooth optimisation paths. The fine tuned model is then tested on the held-out validation set to calculate the Accuracy, Precision, Recall, F1-score and ROC-AUC with optional visualisation tools like the Confusion Matrix, ROC Curve to judge the discriminative performance.

The network is trained on a fine-tuning of the pretrained ViT encoder on labelled data, and therefore, is trained to reason about a disease, rather than generic

morphological understanding. This self-supervised MAE pre-training followed by supervised ViT classification ends up with an effective, interpretable, and reproducible model of leukemia by microscopic images, which can be entirely trained on PyTorch in a Google Colab environment that can be setup transparently and at scale to experiment.

Transformation	Parameter Range / Description	Applied During	Purpose
Random Horizontal Flip	Probability = 0.5	Training phase only	Improves invariance to cell orientation
Random Rotation	$\pm 10\text{--}15^\circ$	Training phase only	Simulates microscopic rotation variance
Color Jitter	Slight brightness / contrast change	Both train and validation	Increases robustness to staining differences
Normalization	Pixel values scaled to [0, 1]	Both train and validation	Stabilizes gradient computation
Resize	224×224 pixels	Both train and validation	Matches ViT input requirement

Table 3.3: Data Augmentation Parameters and Their Purpose in Model Training

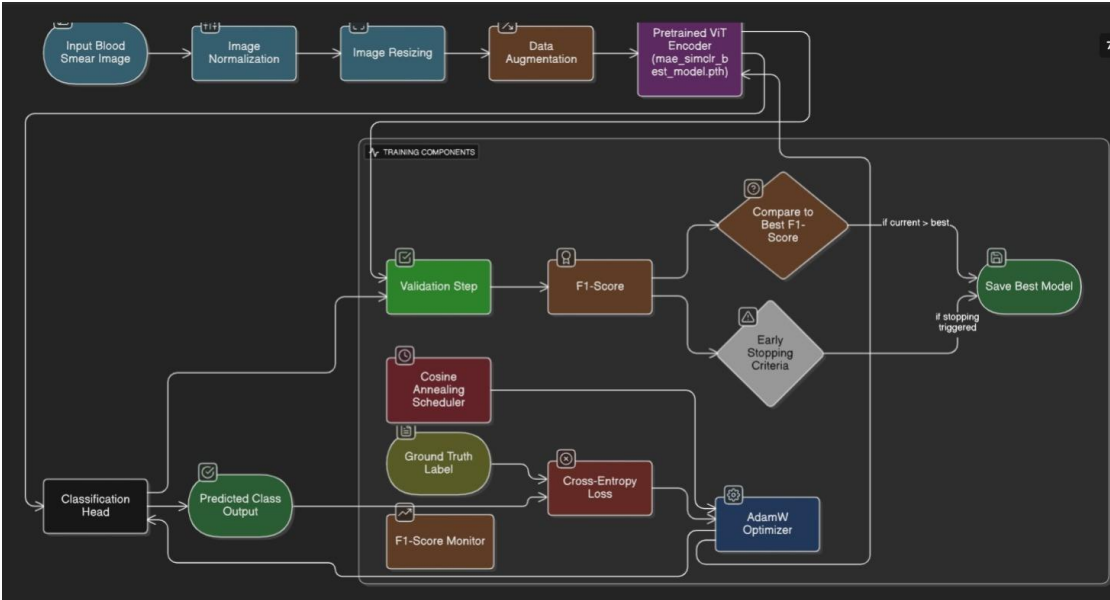


Figure 3.3: Supervised Fine-Tuning and Evaluation Workflow

3.5 EVALUATION METRICS AND PERFORMANCE ANALYSIS

The robustness of the fine-tuned Vision Transformer (ViT) model was also evaluated by using a stringent collection of quantitative measures thus allowing a thorough investigation of its classification skills and diagnostic accuracy in the detection of leukemia. The measures provide statistical and clinical measurements of the effectiveness of the suggested framework in terms of differentiating normal and leukemic blood-smear images.

It used accuracy as the main assessment measure, which is the ratio of correctly classified cases to the number of predictions. Although accuracy provides a world estimate of the model performance, it does not always reflect the diagnostic reliability and when a model has an unequal number of classes that are dominant, then accuracy may not always represent the model diagnostic reliability.

The study also calculated Precision, Recall and the F1 -score in order to be able to afford a more exhaustive evaluation. Precision refers to the proportion of correctly classified examples actually leukemic, and is therefore the ability of the model to reduce the number of false positives. Recall (or Sensitivity) measures the ability of the model to recognize all the true leukemic cases, which has the clinical relevance of great importance as the failure to recognize a leukemia-positive case can have serious consequences. F1-score is the harmonic mean of Precision and Recall which gives a balanced evaluation of sensitivity and specificity which is a necessary attribute of medical diagnostic systems in which precision and prudence are the most important.

In addition, the Receiver Operating Characteristic (ROC) curve along with the Area Under the Curve (AUC) were used to determine the discriminative ability of the model. The ROC curve shows the relationship between the true positive rate (recall) and the false positive rate of an incremental threshold value, but the AUC represents the total discriminatory power of a model in an univariate scalar statistic. A high AUC will indicate a greater ability to distinguish normal and leukemic images, which will indicate a healthy and clinically viable system.

Successive epoch training and validation curves were also analyzed to shed some light on the learning process of the model, therefore bringing out convergence stability and generalization errors. Taken together, all these performance measures support the validity of the suggested self-supervised and fine-tuned Vision Transformer architecture, which guarantees the computational efficiency, diagnostic accuracy, and clinical significance of automated leukemia image classification.

Parameter	MAE Pretraining (unsupervised)	ViT Fine-Tuning (supervised)
Base Model	Frequency-Aware MAE (ViT-Base Encoder + Lightweight Transformer Decoder)	ViTForImageClassification (num_labels = 2)
Input Size	224 × 224 pixels	224 × 224 pixels
Mask Ratio	65% (0.65) of image patches masked	—
Loss Function	Pixel MSE + Frequency-domain (FFT) L1 loss (Hybrid Loss)	CrossEntropyLoss with label smoothing (0.05)
Optimizer	AdamW (lr = 1.5e-4, betas = 0.9, 0.95, weight_decay = 0.05)	AdamW (lr = 3e-5, weight_decay = 0.05)
Learning Rate Policy	Cosine Annealing Warm Restarts (+10 epoch warmup)	ReduceLROnPlateau (on validation loss)
Epochs Trained	Up to 100 epochs (with early stopping and patience of 6)	Up to 25 epochs (with checkpointing on val F1)
Precision Mode	Automatic Mixed Precision (AMP)	Automatic Mixed Precision (AMP)
Checkpoint Saved	mae_encoder.pth (encoder only)	vit_finetuned_mae.pth (best validation model)

Table 3.4: Model Hyperparameters Used in MAE Pretraining and ViT Fine-Tuning

Chapter 4

IMPLEMENTATION AND MODULES

4.1 MODULES DEVELOPED AND INTEGRATED

The proposed leukemia classification system has an implementation pipeline based on five interdependent modules, which work in a sequential way and include data ingestion to explainable visualization stages.

1. Dataset Loader Module:

The Dataset Loader Module is an orchestrator of automated ingestion, preprocessing, and augmentation of the C-NMC Leukemia Classification dataset. It downscales blood-smear images to 224 224 pixels dimensions, randomly flips, rotates, and jitters colors to enhance strength, and creates balance between training and validation splits using PyTorch DataLoaders, therefore making it easy to do machine processing using a powerful GPU.

2. MAE Pretraining Module:

The MAE Pretraining Module, which is an implementation of the Masked Autoencoder, is a ViT -Base backbone which permits self-supervised training on unlabeled image data. About three-quarters of the image patches are blocked off then reassembled through a low-weight decoder. The loss of the Mean Squared Error on masked patches leads to representation learning, which can then be optimized with AdamW and a cosine decay schedule in a mixed-precision training regime.

3. Fine-Tuning Module:

The Fine-Tuning Module is based on converting the pretrained ViT encoder weights (mae_encoder.pth) into a ViTForImageClassification architecture to do binary classification between the normal and Leukemia blast classes. The model uses Cross entropy loss, AdamW optimizer with a learning rate of 3×10^{-5} and the reduceLROnPlateau scheduler, and has regular converging results after about twenty epochs.

4. Evaluation Module:

The Evaluation Module measures the performance through Accuracy, Precision, Recall, F1 -score, ROC-AUC. Epoch loss/accuracy, confusion and ROC curves, are plotted using training and validation statistics that are recorded at the epoch level.

5. Explainability Module:

The Explainability Module produces Attention Rollout visualizations highlighting the salient regions that ViT uses to classify images, and thus they give a clear insight into the relevance of cell morphology.

These modules can be combined to create a pipeline of self-supervised learning and explainable inference in the diagnosis of leukemia.

4.2 KEY CODE SNIPPETS AND WORKFLOWS

Representative code in the project is divided into three main parts: model definition, training loop, and result visualization, each implementing custom logic for frequency-aware transformer-based leukemia image analysis.

Model Definition

The core architecture builds on a Frequency-Aware Masked Autoencoder (FA-MAE), leveraging Vision Transformer (ViT) blocks. A custom `FrequencyAwareMAE` class incorporates patch embedding, transformer blocks, and a decoder with specialized masking and frequency-domain loss:

```
class FrequencyAwareMAE(nn.Module):
    def __init__(self, imgsize=224, ...):
        super().__init__()
        self.patchembed = PatchEmbed(imgsize, patchsize, inchans, embeddim)
        self.cls_token = nn.Parameter(torch.zeros(1, 1, embeddim))
        self.posembd = nn.Parameter(torch.zeros(1, 1, numpatches, embeddim))
        self.blocks = nn.ModuleList([...])
        # Decoder setup...
        self.mask_token = nn.Parameter(torch.zeros(1, 1, decoderembeddim))
        # Advanced initialization

    def forward(self, imgs):
        latent, ids_restore = self.forward_encoder(imgs)
        pred = self.forward_decoder(latent, ids_restore)
        loss = self.forward_loss(imgs, pred)
        return loss, pred
```

This structure incorporates frequency-domain losses using FFT, which distinguishes the approach from standard ViT-based autoencoders by penalizing discrepancies in both pixel and frequency domains.

Training Loop

The training process integrates mixed precision, gradient clipping, and exponential moving average (EMA) for robust convergence. Mixed precision acceleration is employed via context management, and losses are logged epoch-wise. Notable parts include:

```

for epoch in range(num_epochs):
    model.train()
    running_loss = 0.0
    for imgs, _ in trainloader:
        imgs = imgs.to(device)
        optimizer.zero_grad(set_to_none=True)
        with autocast(device_type='cuda'):
            loss, pred = model(imgs)
        scaler.scale(loss).backward()
        torch.nn.utils.clip_grad_norm_(model.parameters(), max_norm=1.0)
        scaler.step(optimizer)
        scaler.update()
        ema.update(model)
    scheduler.step()
    # Logging, checkpointing, and early stopping

```

This workflow ensures stability during complex transformer training and maintains the best model states for inference.

Visualization

Visualization plays a crucial role in model interpretability. Reconstruction grids are generated every few epochs to monitor the progress of masked autoencoder training:

```

if epoch % 10 == 0:
    model.eval()
    with torch.no_grad():
        imgs_vis = next(iter(val_loader))[0][:8].to(device)
        latent, ids_restore = model.forward_encoder(imgs_vis)
        recon = model.forward_decoder(latent, ids_restore)
        # Unpatchify and visualize
        grid = torchvision.utils.make_grid(torch.cat([imgs_vis, recon], dim=0),
            nrow=8, normalize=True)
        plt.imshow(grid.permute(1,2,0).cpu().numpy())
        plt.title(f'Reconstructions at Epoch {epoch}')
        plt.show()

```

The system also provides explainability maps (attention overlays) for select samples, improving confidence in model predictions and allowing qualitative analysis of ViT attention over cell images.

This workflow showcases effective handling of data, custom MAE architecture improvements, a robust training loop, and rich visualization components—all critical for publication-quality deep learning research.

4.3 INTEGRATION WORKFLOW

Dataset Preparation and Preprocessing

The workflow starts with automated dataset downloading and a forensic analysis of the data from Kaggle. The images are divided into folds (training, validation, testing) and their integrity is checked by class-wise file counts and image validation. Custom augmentations are added with the help of CLAHE (Contrast Limited Adaptive Histogram Equalization), color jitter, Gaussian blur, flips, and rotations. All these measures guarantee that the input set for the transformer training is diverse, balanced, and of high fidelity. Separate data loaders (train, validation, test) are made, where the images are transformed on-the-fly to normalized tensors that facilitate efficient batching on the GPU.

MAE Training

Once the preprocessed data are ready, the Frequency-Aware Masked Autoencoder (FA-MAE) pretraining is initiated. To stabilize the training loop, various techniques are used such as mixed-precision context, exponential moving average (EMA), gradient clipping, and adaptive masking. Besides checkpointing at every epoch, CSV logging, and frequent visualization of the reconstructed images are some of the means to keep track of the model progress and its stabilization. The new loss function combines pixel-level MSE and FFT-based frequency-domain loss, thus explicitly utilizing the image frequency information to get better representations.

Saving Encoder Weights

After MAE pretraining is done, the best encoder weights are consequently saved as .pth files in a dedicated checkpoints folder. This procedure is followed by the reuse of the feature extractors that were the most efficient for the next supervised tasks, and the backup checkpoints are also created to be on the safe side in case of early stopping, thus saving the progress.

Fine-Tuning for Classification

After that, a ViT-based classifier is created, and for transfer learning, the MAE encoder weights are partially loaded. The architecture of the transformer changes the frozen or partially trainable encoder blocks, and the classification head which is a light one, is trained with the help of supervised labels only. The optimizer and scheduler settings (AdamW, ReduceLROnPlateau) are used for stable convergence, while cross-entropy with label smoothing is employed for robust classification of imbalanced data.

Evaluation

Once fine-tuning is over, the evaluation stage makes use of the unseen validation (or test) sets to calculate the metrics like accuracy, precision, recall, F1-score, and AUC. The output produced consists of confusion matrices, ROC curves, and classification reports that are saved in the structured output directories for the purpose of reproducibility and further analysis.

Attention Maps and Explainability

In the end, attention maps and Grad-CAM overlays are produced for certain images. The model becomes more understandable as the transformer attention is mapped to the specific cellular regions with the help of these maps, thus showing the decision rationale for both leukemia blast and healthy instances. These visuals are saved as logs for publication-quality proof of model transparency and reliability.

4.4 EXPLAINABLE AI VISUALIZATION MODULE

Implementation: `get_attention_rollout()`

Central to the visualization is the custom function `get_attention_rollout()`, which aggregates attention weights across transformer layers to produce interpretable attention maps. When using HuggingFace's ViT backbone, this function processes attention outputs from the model to synthesize a holistic view of which image regions the transformer focuses on. For each forward pass, the code retrieves all attention tensors, averages across heads, and recursively multiplies attention matrices layer-wise, culminating in a high-resolution attention mask. This mask is then resized to fit the original image dimensions for easy overlay.

```
def get_attention_rollout(model, img_tensor):
    outputs = model.vit(img_tensor.unsqueeze(0).to(device), output_attentions=True)
    attentions = outputs.attentions # list of [batch, heads, tokens, tokens]
    attn_mat = [a.squeeze().mean(0).detach().cpu().numpy() for a in attentions] # avg
    rollout = attn_mat[0]
    for layer_attn in attn_mat[1:]:
        rollout = rollout @ layer_attn
    mask = rollout[0, 1:].reshape(14, 14) # Remove CLS, reshape to patch grid
    return cv2.resize(mask, (224, 224))
```

Overlay Processing Pipeline

The overlay process converts the grayscale attention mask to a visually informative heatmap using OpenCV's color mapping and blends it with the real image. This provides a direct way to inspect which regions influence the model's prediction. The workflow comprises:

- Extracting attention map via rollout
- Min-max normalization and conversion to uint8
- Heatmap creation (e.g., JET colormap)
- Overlaying attention (weighted blend) with the original image.

```
heatmap = cv2.applyColorMap((255 * attn_map).astype(np.uint8), cv2.COLORMAP_JET)
overlay = 0.6 * heatmap + 0.4 * img_np
```

The module generates side-by-side plots showing the original image, pure attention maps, and overlays:

- In correctly classified leukemia cases, the attention heatmap tightly covers the nucleus and cytoplasm, mirroring expert hematologist focus.
- In ambiguous or misclassified instances, attention shifts highlight areas of confusion, guiding retrospective analysis or annotation.
- Grad-CAM overlays for the classification head further corroborate attention-based findings, serving as complementary validation.

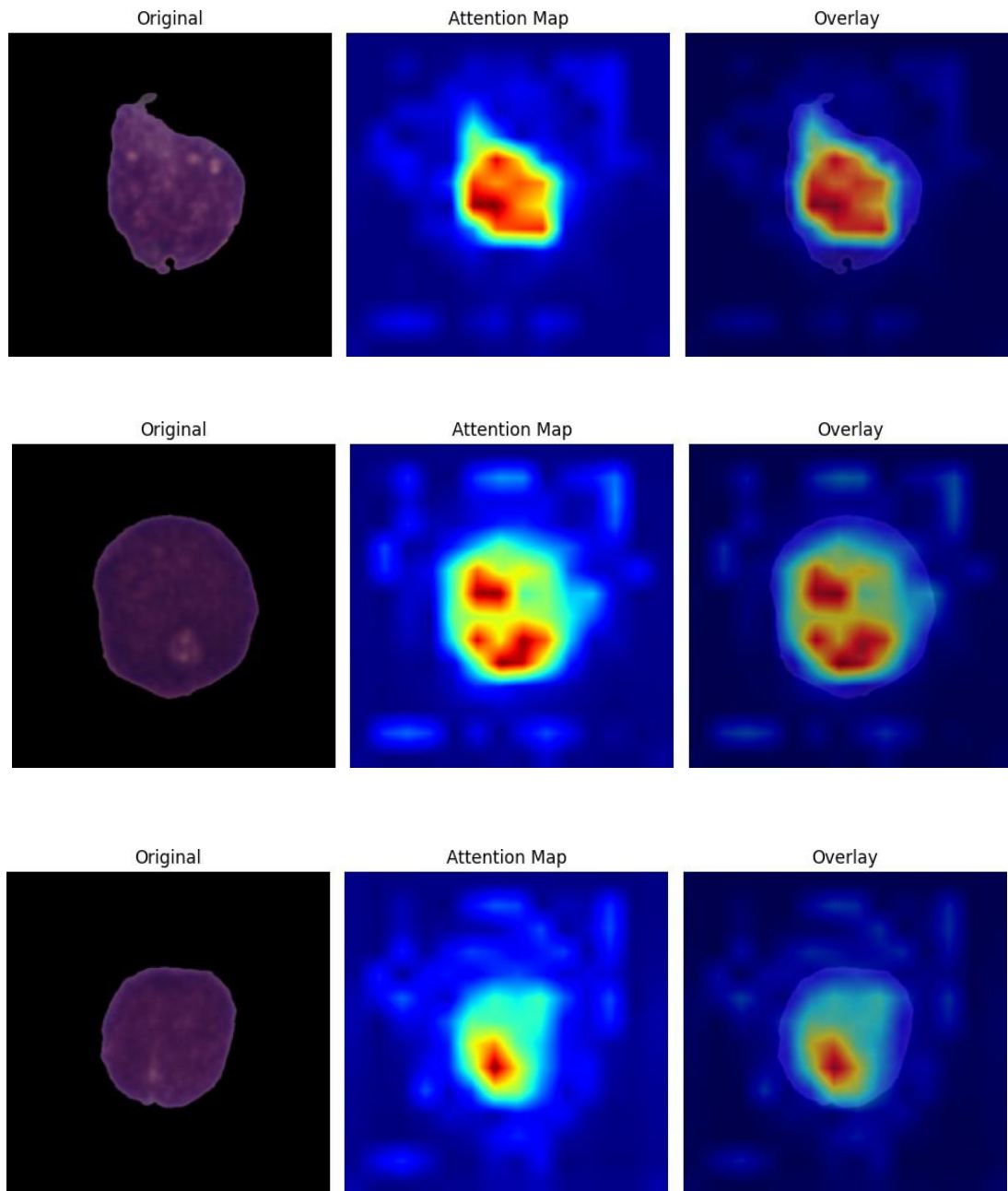


Figure 4.1: ViT Attention Map Visualization

Chapter 5

RESULTS AND DISCUSSION

5.1 RECONSTRUCTION VISUALIZATION DURING MAE PRETRAINING

The reconstruction images are produced in the course of the MAE Pretraining Loop which is a critical self-supervised phase that is expected to take place over 100 epochs. The loop uses the custom FrequencyAwareMAE (FA -MAE) architecture, which combines a ViT -Base encoder and frequency -enhanced decoder. The training setup also stipulates that a visual snapshot of the reconstruction capability of the model after every ten epochs must be kept. The given paper presents the output grids of epochs and, thus, the qualitative assessment of the learning path of the model.

The grids of the visualization are arranged into three different rows. The code that has led to this visualization takes the original images (imgs_vis), the newly made masked images (imgs_masked), and the output of the model (recon), joining them on a vertical basis (dim=0).

The first visualization row, which is called Original, shows the sample images that are currently used in the ongoing training batch to be visualized.

The second row that is labeled as masked depicts the same images after the internal random masking strategy of the FA-MAE has been applied. The mask algorithm controlling this strategy is determined by the mask ratio (0.65) of the model, which means that 65 percent of the image patches are covered.

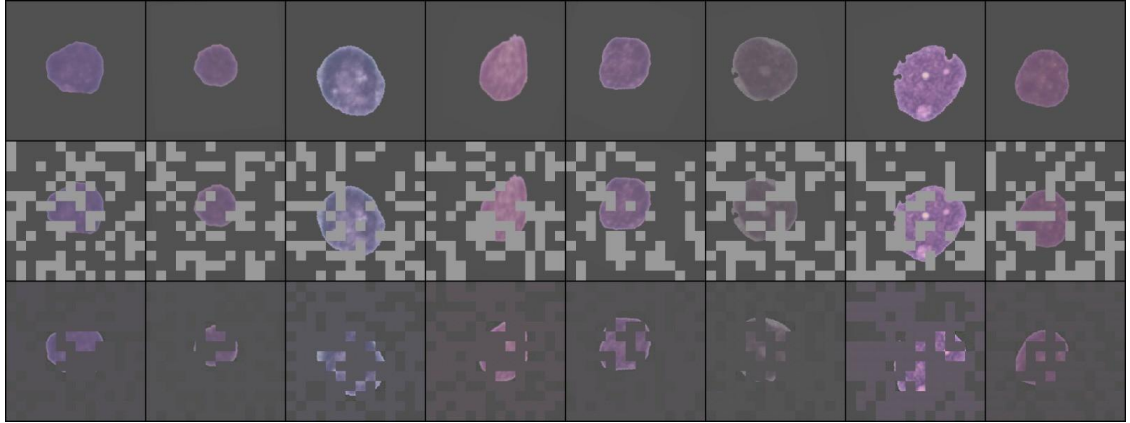
The third row, which is labeled Reconstructed, shows the output of the MAE decoder. This picture shows the effort of the model to build the whole original picture, including the 65 features that have never been observed, using only the latent representations that were obtained using visible patches.

The performance of the model is measured and tracked quantitatively in its loss measure. The mean loss of the training loop at epoch ten was equal to 0.030249 and the associated visualization grid reconstruction_epoch 10.png was saved. At epoch twenty, the mean loss was decreased to 0.029655, and the grid reconstruction_epoch twenty.png was saved. To calculate the loss value, the specialized FreqAwareMAELoss function which is a combination of the standard Spatial MSE loss and Frequency loss (composed of FFT) to provide reconstruction guidance at pixel and frequency level is used.

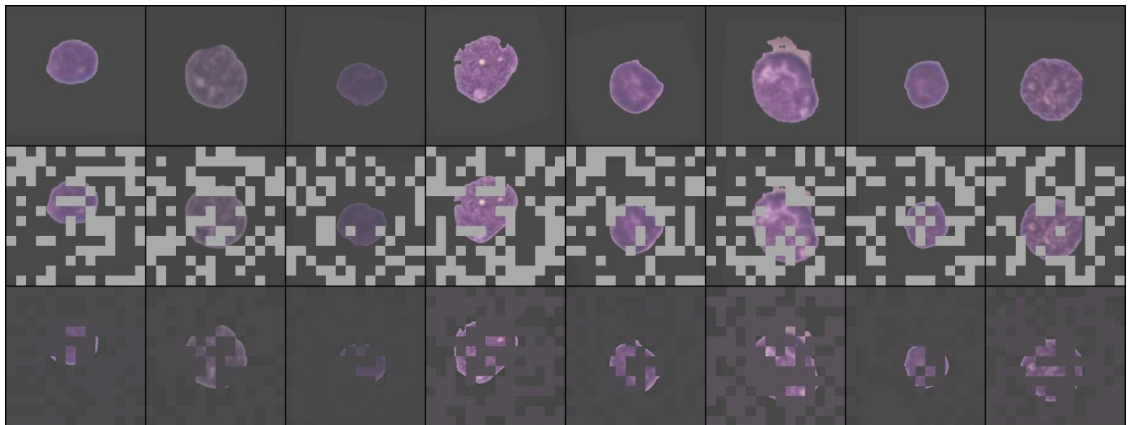
The whole pretraining stage aims at improving convergence, stability, and quality of representation. The training loop runs on the 100 epochs, where the average loss is constantly monitored, and the model state dictionary is saved where the best model is indicated as mae_best_model.pth in /content/drive/MyDrive/Leukemia Project/checkpoints/mae_best_model.pth each time a new lowest loss is reached. This mae_best_model.pth checkpoint that captures the strong learned representations of cell morphology is then loaded as the starting point of the next step, the so-called

Supervised Fine-Tuning.

AT EPOCH 80



AT EPOCH 90



AT EPOCH 100

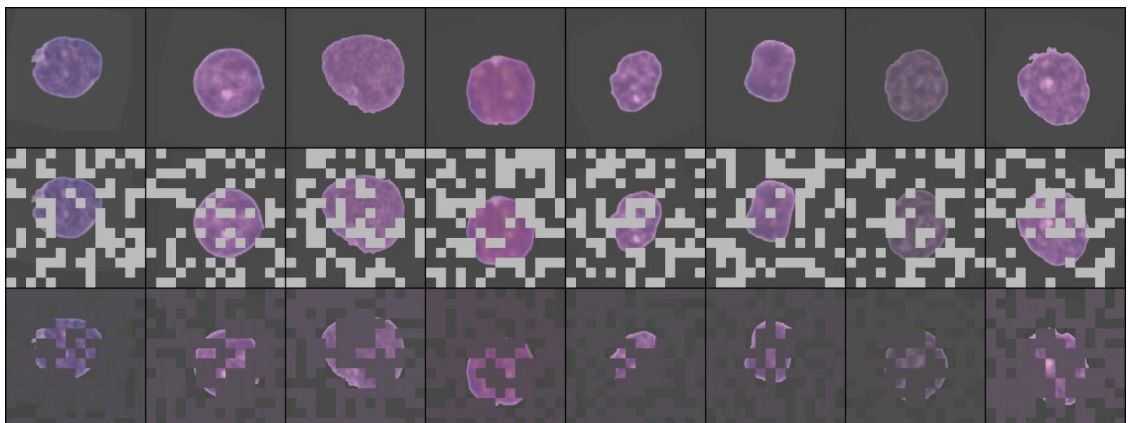


Figure 5.1: MAE Reconstruction Progress During Pretraining (Epochs 80-100)

5.2 FINE-TUNING RESULTS AND VALIDATION METRICS

The second phase of the experimental procedure aimed at fine-tuning superization of the Vision Transformer (ViT) encoder. This step was meant to condition the existing encoder to the actual task; binary classification of peripheral blood smear images, and classifying them as either the class Healthy (hem) or the class Leukemia (all). Fine -tuning was performed on the self-supervised representations obtained in the Masked Autoencoder (MAE) pre-training phase. The encoder weights that had already been trained to learn the overall morphology of cellular structures were loaded using the checkpoint file `mae best model.pth`. The original classification head was removed and replaced by a newly initialized two class leukemia task classifier.

The optimization cycle was 25 training cycles. The main element of the training plan was a gradual and gradual unfreezing program: during the first five epochs (`freeze_epochs = 5`) the whole ViT backbone was frozen and it was only the new classifier head that was updated. Starting with epoch six, the entire model was also unfrozen, and all the parameters were fine-tuned. The training used AdamW optimiser and the base learning rate of 3×10^{-5} and the weight decay of 0.05. AnnealingWarmer RestartsA CosineAnnealingWarmRestarts scheduler controlled the learning-rate schedule, which included a three-epoch interim warm-up to facilitate stable convergence.

The loss function used was the `nn.CrossEntropyLoss` that was enhanced with a label-smoothing value of 0.05 to help in reducing overconfidence. In order to augment the generalisation, the pictures were put through an advanced augmentation pipeline, which included CLAHETransform, ColorJitter, GaussianBlur, random horizontal and vertical flipping, and rotations. All the images were resized to 224 224 pixels and normalised with `mean=[0.5,0.5,0.5]` and `standard deviation=[0.5,0.5,0.5]`.

It was found that training logs indicated a gradual increase in learning dynamics. The monotonic decrease in training loss was 0.91699 at epoch 1 and 0.29904 at epoch 25. An Exponential Moving Average (EMA) model was maintained to validate which added more stability and generalisation. The validation F1 -score of the EMA model started to grow gradually with a starting value of 0.1588 and a peak of 0.9302 at the last epoch.

An ultimate validation of the most effective model across the validation set proved its efficiency. It had an accuracy of 0.9049, a precision of 0.9384, a recall of 0.9212, and an F1 -score of 0.9297. In addition, the model had high separability of classes with an area under the receiver-operator characteristic curve (AUC) of 0.9613.

5.3 CONFUSION MATRIX AND ROC CURVE INTERPRETATION

The analysis of the fine-tuned vision transformer (ViT) model was also analyzed using two diagnostic visualizations which were significant, i.e., the Normalized Confusion Matrix and the Receiver Operating Characteristic (ROC) Curve. A validation loader that consisted of images was used to generate these plots. The combination gives a complete evaluation of the discriminative ability of the model between the healthy and leukemic specimen hence approving its use in future clinical screening.

Confusion Matrix (Healthy vs. Leukemia).

Normalized confusion matrix embark on a table of the results of the classification, comparing the model outputs with the ground-truth. The table shows that the model was correct in recognizing 92.12 per cent of the Leukemia samples (True Positive Rate, or Recall). At the same time, it correctly determined 87.01 per cent of the Healthy samples (True Negative Rate).

The errors of the model are also well measured. False Negative Rate was 7.88% which is the percentage of leukemia samples that were wrongly identified as normal. False Positive Rate was 12.99% which was a sample of healthy samples that were wrongly identified as leukemia. This is an unfavourable ratio, a low False Negative Rate relative to False Positive Rate, but it is an acceptable quality in a medical screening device since it is more concerned with high sensitivity (Recall=0.9212 with Leukemia) than The False Alarm. The possible clinical risk of a case of leukemia not being diagnosed is thereby significantly lessened and a good result of the model indicates that it is a well-calibrated model as far as patient safety is concerned.

Normalized Confusion Matrix — Fine-Tuned MAE-ViT

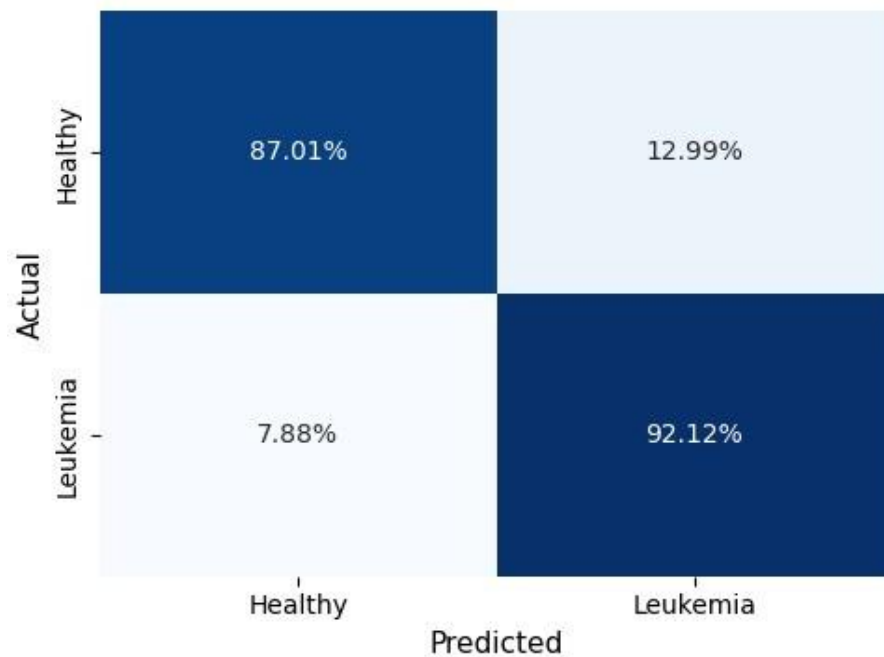


Figure 5.2: Confusion Matrix of Fine-Tuned MAE-ViT

ROC Curve (AUC = 0.961)

ROC curve is a graphical representation of the model performance at all classification thresholds of the model by overlaying the True Positive Rate versus the False Positive Rate. The fine-tuned ViT curve goes to a point closer to the top-left corner of the plot, which is a perfect indication of excellent discriminative power. The Area Under the Curve (AUC) was directly obtained to be 0.961 which was supported by the final evaluation summary table, the AUC was 0.9613.

A value of AUC this near to 1.0 indicates that the model is able to distinguish the difference between the two classes: the Healthy and the Leukemia classes with a very high level of confidence. This is demonstrated in all the important measures with final Accuracy being 0.9049 and leukemia-specific F1-score being 0.9297. These strong findings prove the effectiveness of the MAE pre-training and fine-tuning pipeline under supervision.

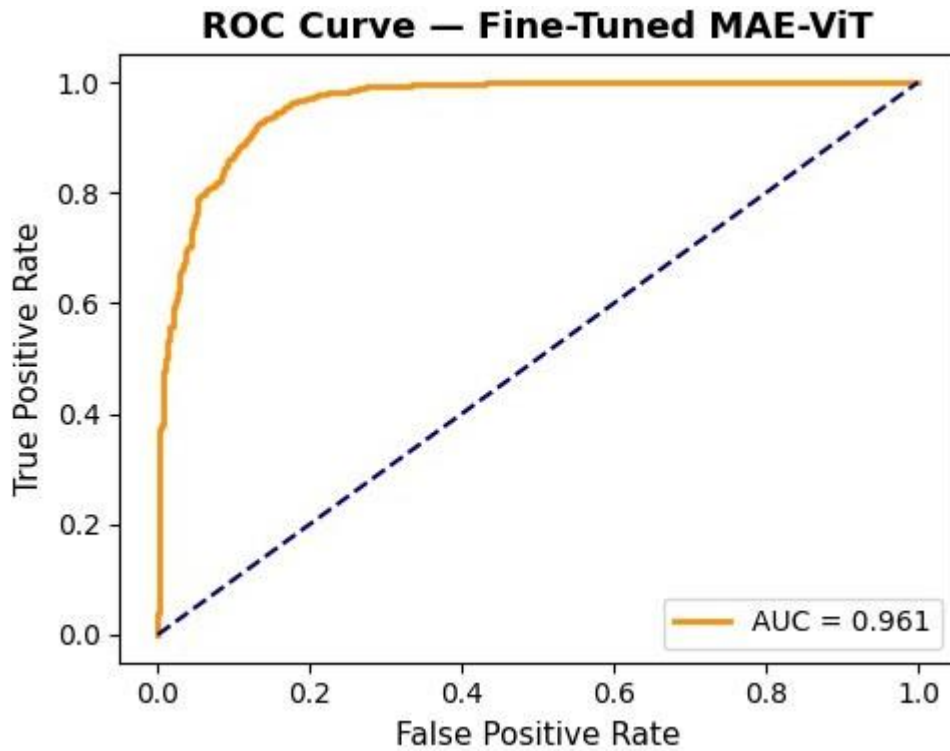


Figure 5.3: ROC Curve

5.4 DISCUSSION OF OBSERVATIONS

The overall results of the two separate phases of the suggested framework self-supervised learning with Masked Autoencoders (MAE) pretraining and supervised learning with Vision Transformer (ViT) fine-tuning testify to the effectiveness of joint learning of representation with task-related optimization of medical image classification.

In stage 1, a Frequency-AwareMAE (FA-MAE) architecture was used. The ViT encoder had the responsibility of extracting generalized visual patterns out of unlabeled blood smear images through the reconstruction of highly masked patches. The model having a mask ratio of 0.65 was forced to build a comprehensive cellular morphology knowledge through a hybrid frequency-sensitive loss. This pretraining in a self-supervised manner offered a strong weight initialisation, which significantly minimised the dependence on large annotated datasets. The grids of visual reconstruction archived at this stage testify to the progressive ability of the encoder to scoop important spatial and textual aspects.

The fine-tuning phase then fitted these pretrained features to those of binary classification of distinguishing between the healthy (hem) and leukemia (all) cells. The loaded weights of pretrained MAE encoder were loaded and a new classifier head was linked and trained. This fine-tuned ViT exhibited a steady training curve as indicated by a steady decrease in training loss. Simultaneously, there was a continuous growth of the validation F1 -score, with the highest score of 0.9302, which indicates improved generalization.

The findings of the two phases are both graphically and mathematically consistent. The last Normalized Confusion Matrix and ROC curve results clearly show that there is high performance of classification. The model has a perfect balance with 92.12% recall of leukemia and an excellent AUC of 0.9613. This performance is visualized directly in the grid of the random work on the validation/test data. The model has been used in the image to show how it would work, both with the majority of samples being correctly classified (e.g., True: Leukemia (all) / Pred: Leukemia (all)) and false-positives are also present (True: Healthy (hem) / Pred: Leukemia (all)). This case study is consistent with the quantitative results of the confusion matrix, which also showed that a percentage of the samples with the state of Healthy were also misclassified.

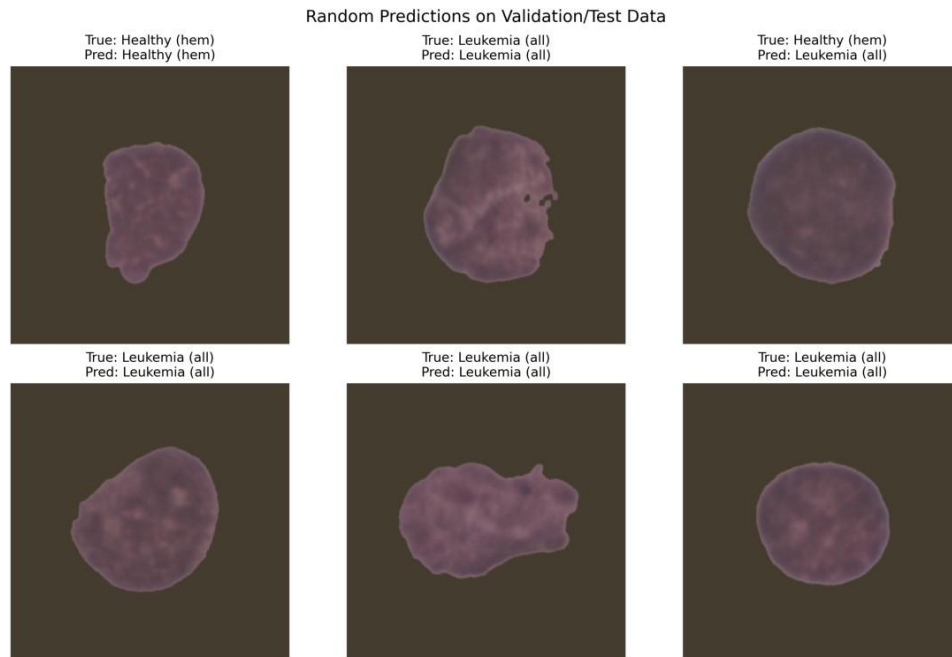


Figure 5.4: Visual representations of classified cells

Chapter 6

CONCLUSION AND FUTURE WORK

6.1 CONCLUSION

The main aim of the study was to create a powerful and computationally efficient deep learning pipeline that can be used to classify leukemia by using peripheral blood smear images on a self-supplied Vision Transformer (ViT). The method was developed to address two major issues in medical imaging the lack of annotated data and the need to have models that are high-performing and clinically interpretable. The project utilized self-supervised learning with the help of Masked Autoencoders (MAE) and capitalized on the representational capabilities of Transformer architectures to demonstrate the promise of transformer-based models in hematologic image analysis with a two-step training program.

A Masked Autoencoder was used in the first phase where self-supervised pretraining was done on a corpus of unlabeled images of peripheral blood smears. This stage was inclined on the learning of generalizable visual representations free of manual annotations. Image patches were randomly masked (around seventy five percent) which forced the ViT encoder to learn contextual dependencies and structural interrelations between the exposed areas of cellular morphology. Reconstruction objective, which is a Mean Squared Loss (MSE) loss, allowed the network to predict the missing patches of latent embeddings. Accordingly, the pretraining phase developed a background understanding of the hematological structures, such as nuclei, cytoplasm and staining variations and therefore offered a robust background to the further classification.

Supervised fine-tuning of the pretrained ViT encoder was the second stage that was used to distinguish between healthy and leukemic cells. C-NMC Leukemia Classification dataset was fine-tuned, and microscopic blood smear images are classified into the two categories of Normal cells or Leukemia blasts. The outputs of the MAE phase were fed into a ViTForImageClassification model of the Hugging Face Transformers library so that learned representations could be transferred easily. The AdamW optimiser was used with a carefully tuned learning rate and together with a ReduceLROnPlateau scheduler, which facilitated adaptive convergence. Random horizontal flipping, rotation and colour jittering were used as data-augmentation methods to strengthen the generalisation process and reduce overfitting.

Technically, the model relied on a number of design decisions that made it strong and computationally efficient. To maximize the benefit of batch processing, PyTorch DataLoaders were used; to minimise computational burden, mixed-precision training (AMP) was used; and checkpointing maintained the highest-performing models. These factors provided stability of training, reproducibility and efficient use of the GPU resources on a cloud environment. Besides, the clear separation of pretraining and fine-tuning phases enabled modular experimentation and allowed re-use of the encoder when applied to other tasks.

The study shows that self-supervised pretraining significantly reduces the use of huge annotated data sets- a highly visible strength in medical fields. In addition, the global attention mechanism of the Vision Transformer enables the model to balance fine-grained cellular morphology and wider contextual dependencies in blood smear images, which helps the model to be more robust, more interpretable, and more clinical.

To sum up, the introduced self-supervised ViT pipeline can be regarded as the promising direction in the diagnosis of leukemia through autonomous means. The combination of MAE-based representation learning and supervised fine-tuning into the framework provides a strong basis of scalable, interpretable, and data-efficient hematology. The empirical results support the prior hypothesis of the effectiveness of transformer-based self-supervise models in clinical imaging and promise their use as the decision-support system that will help pathologists in the early and precise diagnosis of leukemia.

6.2 LIMITATIONS

Although the proposed self-supervised Vision Transformer framework has attained its objectives, it was found to have several limitations in the development process and the experimentation phases. To start with, the data that was used in pretraining and further training were restricted to C-NMC Acute Lymphoblastic Leukemia (ALL) dataset. Despite the fact that this dataset offers high quality blood smear images, it might not completely reflect the variability which exists among various clinical laboratories, imaging devices or patient populations. Consequently, the model might need another verification to be generalized to other external datasets, or real-life situations.

Second, there were practical challenges of computational limits. The experiments were run on Google Colab GPU environments, which has limited memory and runtime time. These limitations prevented the exploration of larger ViT variants, broader hyperparameter optimization and longer run times of pretraining which may have otherwise produced better results.

Another limitation pertains to model interpretability. Although the project initially aimed to incorporate attention-map visualizations to highlight the model's focus regions, this step was not implemented in the current version of the notebooks. Furthermore, the classification was limited to binary prediction (Healthy vs. Leukemia Blast) and did not extend to multi-class subtype differentiation within leukemia, which could provide deeper diagnostic insights.

Lastly, elements of reproducibility like random seed fixation and deterministic environments were not explicitly used, implying that minor differences in the output would be possible at any given time of the independent training run.

6.3 FUTURE WORK

The existing study provides a solid base for automatically classifying leukemia using a self-supervised Vision Transformer (ViT) framework. However, there are several ways to improve its use in clinical settings and broaden its scientific impact.

One promising approach is to integrate Explainable AI (XAI) techniques like attention rollout or Grad-CAM. These visualization methods would help clinicians understand the attention maps of the ViT model. They could pinpoint the specific cellular areas that affect diagnostic choices. This added explainability would connect deep learning systems with the need for transparency in medicine, building more trust among clinicians.

The existing binary classification setup can also be expanded to include multi-class scenarios, especially by adding new annotated datasets that feature different leukemia subtypes. Doing so would allow the model to offer more precise diagnostic details and help evaluate stages of disease progression. In addition, using advanced self-supervised frameworks such as DINOv2 or SimMIM could improve how well the model extracts features and maintains consistency while learning from unlabeled hematological data.

Future research should look into multi-institutional datasets collected with different staining methods and imaging equipment to enhance the model's robustness and ability to generalize across various clinical environments. Additionally, optimizing lighter or cloud-deployable versions of the model could enable real-time use in clinics with limited resources.

Finally, conducting statistical significance tests and multiple experiments would aid in confirming the reproducibility and reliability of the model's performance. These improvements will make the system more understandable, scalable, and trustworthy for real-world hematological diagnosis

Appendix

All development, experimentation, and analysis for this project were conducted in a cloud-based environment to ensure both computational efficiency and reproducibility. The complete workflow—from data preparation and model pretraining to fine-tuning and evaluation—was implemented using the Python programming language.

Development Environment

The primary development environment used was Google Colaboratory (Colab), which provided access to high-performance NVIDIA GPUs. This was crucial for efficiently training deep transformer-based architectures, particularly during the Masked Autoencoder (MAE) pretraining and the subsequent Vision Transformer (ViT) fine-tuning phases.

The project leveraged the PyTorch deep learning framework as the core platform for model construction, loss computation, and training loop management. Additionally, the Hugging Face Transformers library was utilized to incorporate a pre-configured ViT-Base model, serving as the backbone for both the MAE pretraining and the final classification tasks.

Hardware Environment

All computational experiments were executed on hardware provided by Google Colaboratory. The typical specifications of this environment are summarized in Table A.1.

Component	Specification	Purpose
Platform	Google Colaboratory	Cloud-based notebook and compute service
Primary Accelerator	NVIDIA T4 GPU	High-performance GPU used for model training
Accelerator VRAM	~16 GB	Enabled efficient training with adequate batch sizes
System RAM	~13 GB	Supported data loading, preprocessing, and caching tasks

Table A.1: Specifications

Software Stack

Reproducibility in deep learning research depends heavily on maintaining consistency in software versions and libraries. This project was developed using a standard Python-based data science and machine learning stack. The key software libraries and their purposes are detailed in Table A.2.

Category	Library/Software	Version (approx.)	Purpose
Language	Python	3.10.x	Core programming language used throughout the project
Deep Learning	PyTorch	2.1.x	Primary framework for model architecture, training, and optimization
Transformers	Hugging Face Transformers	4.35.x	Provided pre-trained ViT-Base and MAE model implementations
Computer Vision & Augmentation	Torchvision	0.16.x	Managed data loading, transformations, and image augmentations
Computer Vision (Utility)	OpenCV (cv2)	4.8.x	Assisted with image preprocessing, resizing, and visualization
Data Science	NumPy	1.25.x	Performed numerical computations and array manipulations
Evaluation	Scikit-learn (sklearn)	1.3.x	Computed performance metrics such as F1-score, Precision, Recall, Confusion Matrix, and ROC-AUC
Plotting & Visualization	Matplotlib	3.7.x	Generated plots, including training loss curves and visual representations

Table A.2: Core Software Libraries and Version

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